State Emergency Management Plan

Energy Sub-Plan 2023



Photo credit
Iona Gas Underground Storage Facility with Permission from Lochard Energy

Acknowledgment



We acknowledge and respect Victorian Traditional Owners as the original custodians of Victoria's land and waters, their unique ability to care for Country and deep spiritual connection to it. We honour Elders past and present whose knowledge and wisdom has ensured the continuation of culture and traditional practices.

We are committed to genuinely partner, and meaningfully engage, with Victoria's Traditional Owners and Aboriginal communities to support the protection of Country, the maintenance of spiritual and cultural practices and their broader aspirations in the 21st century and beyond.

© The State of Victoria Department of Energy, Environment and Climate Action 2023

This work is licensed under a Creative Commons Attribution 4.0 International licence. You are free to re-use the work under that licence, on the condition that you credit the State of Victoria as author. The licence does not apply to any images, photographs or branding, including the Victorian Coat of Arms, the Victorian Government logo and the Department of Energy, Environment and Climate Change (DEECA) logo. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/

Disclaimer
This publication may be of assistance to you but the State of Victoria and its employees do not guarantee that the publication is without flaw of any kind or is wholly appropriate for your particular purposes and therefore disclaims all liability for any error, loss or other consequence which may arise from you relying on any information in this publication.

Accessibility
If you would like to receive this publication in an alternative format, please telephone the DEECA Customer Service Centre on 136186, email customer.service@delwp.vic.gov.au or via the National Relay Service on 133 677 www.relayservice.com.au. This document is also available on the internet at www.delwp.vic.gov.au.www.deeca.vic.gov.au

Contents

1. Introduction 5
1.1. Purpose 5
1.2. Value 5
1.3. Audience 5
1.4. Scope 5
1.5. Implementation 5
1.6. Authorising Environment 5
1.7. Review 5
1.8. Recognition of Traditional Owners 6
1.9. Linkages and hyperlinks 6
Document 6
Description 6
Joint national and industry plans 6
Related legislations 6
Emergency Management Plans 6
2. Victorian Energy Sector 7
2.1. Electricity 8
2.2. Gas 9
2.3. Liquid Fuels 9
3. Energy Risks and Dependencies 11
3.1. Government assessment and monitoring of risks 11
3.2. Electricity 11
3.3. Gas 14
3.4. Liquid Fuels 15
4. Energy Emergencies 18
4.1. Major Emergencies 18
4.2. Levels of energy emergencies 18
5. Emergency Management Planning 20
State Emergency Management Planning 20
Regional Emergency Management Planning 20
Municipal Emergency Management Planning 20
Critical Infrastructure Planning 20
Sector Resilience Plans 21
Statement of Opportunities 21
Community Emergency Planning 21

OFFICIAL

6. Mitigation 22
6.1. Roles and Responsibilities 22
7. Preparedness 24
7.1. National Energy Objectives 24
7.2. Guaranteed Service Levels 24
7.3. Electricity and Gas Safety 24
7.4. Major Hazard Facilities 25
7.5. Preparedness for weather conditions and natural hazard events 25
8. Readiness 27
8.1. Roles and Responsibilities 27
8.1.1 Electricity 27
8.1.2 Gas 27
8.1.3 Liquid Fuels 27
8.2. Readiness for weather conditions and natural hazard events 28
9. Response (including Relief) 30
9.1. Control arrangements 30
State Controller – Energy 30
9.2. Application of State Emergency Management Priorities 30
9.3. Roles and Responsibilities Energy 32
9.4. State Control Centre – Energy Unit 33
9.5. Concurrent emergencies 34
9.6. Operation of multi-jurisdictional and state-level joint government-industry arrangements 34
9.7. Market Interventions and Legislative Powers 35
9.8. Critical Infrastructure, community assets and essential services 35
9.9. Response to electricity major emergencies 36
Electricity Thresholds and Triggers for Major Emergencies 37
9.10. Mutual Aid Arrangements 39
9.11. Response to gas major emergencies 39
Gas Thresholds and Triggers for Major Emergencies 40
9.12. Life Support Customers (Electricity and Gas) 42
Electricity 42
Gas 42
9.13. Response to liquid fuel major emergencies 42
9.14. Consequence Management 45
9.15. Public Information and Warnings 45
Energy Emergency Advice and Messaging 45

Energy Industry Communications 46
9.16. Relief 46
Transition to Recovery 47
10. Recovery 48
Appendix A: Multi-jurisdictional and State-level joint government-industry arrangements 50
Electricity 50
Gas 51
Liquid Fuels 51
Appendix B: Related plans and arrangements 54
Appendix C: Acronyms 56

List of Tables

TABLE 1: Summary of related plans and arrangements 6
TABLE 2: Electricity Risk Ratings 12
TABLE 3: Gas Risk Ratings 14
TABLE 4: Liquid Fuels Risk Ratings 16
TABLE 5: Summary of triggers, actions, and control status 19
TABLE 6: Resources for Households to Plan for Emergencies 21
TABLE 7: Mitigation measures to risks 22
TABLE 8: Preparedness Arrangements 26
TABLE 9: Readiness Arrangements 28
TABLE 10: State Emergency Management Priorities – Energy 31
TABLE 11: Control arrangements 32
TABLE 12: Lack of Reserve Levels 36
TABLE 13: Electricity Thresholds and Triggers Major Emergency 38
TABLE 14: Gas Major Emergency Thresholds and Triggers 41
TABLE 15: Liquid Fuels Emergency Thresholds and Triggers 44
TABLE 15: Electricity multi-jurisdictional and state-level joint government-industry arrangements 51
TABLE 16: Gas multi-jurisdictional and state-level joint government-industry arrangements 51
TABLE 17: Liquid Fuels multi-jurisdictional and state-level joint government-industry arrangements 53

List of Figures

FIGURE 1: Victorian Transmission Lines 8
FIGURE 2: Victorian Declared Gas Transmission System and Gas Pipelines 9
FIGURE 3: Liquid Fuel Supply Chain 10
FIGURE 4: Example Control structure for concurrent emergencies 34
FIGURE 5: Example government-industry control structure 35

List of Case Studies

BOX 1: IONA Underground Storage Case Study 29
BOX 2: Mallacoota Community Battery Case Study 48

1. Introduction

1.1. Purpose

This Energy Sub-Plan (2023) (sub-plan) is a sub-plan of the State Emergency Management Plan (SEMP) outlining the Victorian arrangements, where applicable, to deliver a coordinated response to energy emergencies.

1.2. Value

This sub-plan provides information about energy supply chains and infrastructure, risks and consequences of disruption, and the arrangements for energy emergency mitigation, preparedness, response and recovery.

This clarity about context, roles and collaboration between agencies enables an integrated and collaborative response to energy emergencies, minimises the likelihood and consequence of these emergencies, and supports community resilience.

1.3. Audience

The audience for this sub-plan is the Victorian Government, agencies, energy industry and other organisations within the Victorian emergency management arrangements that have an interest in the management of energy emergencies. The audience also includes Commonwealth, state and territory governments, and industry stakeholders that have a role in managing energy emergencies.

1.4. Scope

This sub-plan provides strategic information about Victorian preparedness, mitigation, response and recovery arrangements for energy emergencies, with a focus on major emergencies that disrupt electricity, gas or liquid fuel supply.

This sub-plan does not include detail about the operational activities of individual agencies.

This plan does not describe or duplicate arrangements for Class 1 emergencies.

1.5. Implementation

The arrangements in this sub-plan apply on a continual basis and do not require activation.

This sub-plan is intended to support existing operational plans.

1.6. Authorising Environment

The *Emergency Management Act 2013* (EM Act 2013) is the empowering legislation for managing emergencies in Victoria.

The SEMP is prepared under section 60AD of the EM Act 2013. The SEMP outlines the provisions for the mitigation of, response to and recovery from emergencies, and specifies the roles and responsibilities of agencies in relation to emergency management in Victoria.

This sub-plan is a subordinate plan of the SEMP and has been approved by the State Crisis and Resilience Council (SCRC).

1.7. Review

This sub-plan was reviewed and amended following one year of its publication in 2022 in line with s. 60AN of the EM Act 2013.

This sub-plan was current at the time of publication and remains in effect until modified, superseded or withdrawn.

This sub-plan will be exercised and evaluated every three years from the date of publication. Where any emergency management arrangements are reviewed and changed, the sub-plan will be amended, and a revised version issued.

1.8. Recognition of Traditional Owners

The Department of Energy, Environment and Climate Action (DEECA) is committed to genuinely partnering, and meaningfully engaging with Victoria's Traditional Owners and Aboriginal communities to support the protection of Country, the maintenance of spiritual and cultural practices and their broader aspirations.

1.9. Linkages and hyperlinks

This plan refers to a range of existing plans, arrangements and resources relating to energy emergencies. This plan does not seek to duplicate the information contained in these resources, and instead provides links to where the reader can obtain further information (**Appendix B**).

Documents or resources that are referred to frequently throughout this plan, such as the SEMP, are not hyperlinked in each instance.

All hyperlinks were accurate at the time of publication, and the currency of the linked content remains the responsibility of the host agency.

TABLE 1: Summary of related plans and arrangements

Document	Description
Joint national and industry plans	Joint plans and standard operating procedures and governance arrangements to support state and territory government and industry, allowing these agencies to jointly prepare for, monitor and respond to energy incidents and energy emergencies. These include the following: • National Electricity Market Memorandum of Understanding on the Use of Emergency Powers (National Electricity Market Emergency Protocol) • National Electricity Market Emergency Management Forum's Power System Emergency Management Plan • National Gas Emergency Response Advisory Committee's Interruption to Supply Process • Gas Emergency Protocol (2022) • Victorian Electricity Emergency Communication Protocol (2022) • Single Industry Spokesperson Protocol for Electricity in Victoria (2019) • Victorian Gas Emergency Communication Protocol (2019) • Single Industry Spokesperson Protocol for Gas in Victoria (2019) • National Liquid Fuel Emergency Response Protocol • Australian Institute of Petroleum (AIP) Communications Protocol (2015).
Related legislations	The following legislation is directly relevant to the management of Class 2 energy emergencies: • *Liquid Fuels Emergency Act 1984 (Cth)* • *National Electricity Law* • *National Gas (Victoria) Act 2008* • *Electricity Distribution Code of Practice 2022* • *Electricity Industry Act 2000* • *Electricity Safety Act 1998* • *Gas Industry Act 2001* • *Gas Safety Act 1997* • *Pipelines Act 2005*
Emergency Management Plans	Emergency management plans are prepared, maintained and exercised by businesses, the community and infrastructure owners. These entities will activate their plans in response to an energy emergency when pre-determined triggers are met. Further all tiers of government, the EM Act 2013 requires local government and regions to develop and maintain Emergency Management Plans in accordance with the Ministerial Guidelines and which underpin the State Emergency Management Plan.

2. Victorian Energy Sector

The Victorian energy sector includes generation, production, transmission, storage and distribution of energy across three sub-sectors: electricity, gas and liquid fuels. In Victoria, energy sector supply chains and infrastructure are privately owned and operated.

The energy sectors provide essential services to the Victorian community such as lighting, heating and cooling, process heat for manufacturing and transportation of fuel.

The electricity, gas and liquid fuel sub-sectors are highly interconnected and reliant on services they each provide. Each sub-sector also has strong interdependencies with other critical infrastructure sectors and supply chains, both domestically and internationally. This is highlighted by increasingly inter-linked east coast electricity and gas markets, energy assets reliance on reliable water supply and road access, and dependency on international and interstate supplies of natural gas and liquid fuels.

All energy sectors are vulnerable to supply chain disruption and infrastructure damage which can give rise to supply disruption. Risks arise in the context of global and domestic supply chains of products, parts and people. These include interdependent national markets, increasing domestic and international attention on decarbonising the energy economy, and intra-dependencies across the social, built, economic and natural environments.

Victoria has committed to reaching net-zero emissions and a climate-resistant Victoria by 2045. As such, the energy sector emissions reduction pledge accelerates Victoria's transition to a clean and efficient energy future. This transition aims for 50 per cent renewable sources by 2030 within the electricity system. This transition will see:

- large scale solar and wind farms and storage solutions, such as utility-scale batteries
- the retirement of increasingly expensive and unreliable coal-fired power stations as the state moves away from fossil fuels
- the pioneering of innovative energy sources, such as renewable hydrogen and offshore wind
- a wave of electrification with a move away from gas in homes to deliver more climate-friendly energy efficient homes
- new technologies, such as rooftop solar, neighbourhood and household batteries
- new and upgraded energy infrastructure ensuring clean energy for all.

Wind Farms



Credit: David McArthur – Parallax Photography

Solar Farms and Battery



Credit: Edify Energy

Victorian Big Battery



Credit: Neoen 16

Energy distribution, transmission and critical infrastructure owners/operators are an essential part of planning to mitigate the risk and consequences from major emergencies. This planning is essential in providing safe and reliable supplies of electricity, gas and liquid fuels to Victoria and, importantly maintaining supply to other critical infrastructure sectors reliant on energy to provide essential services to the state.

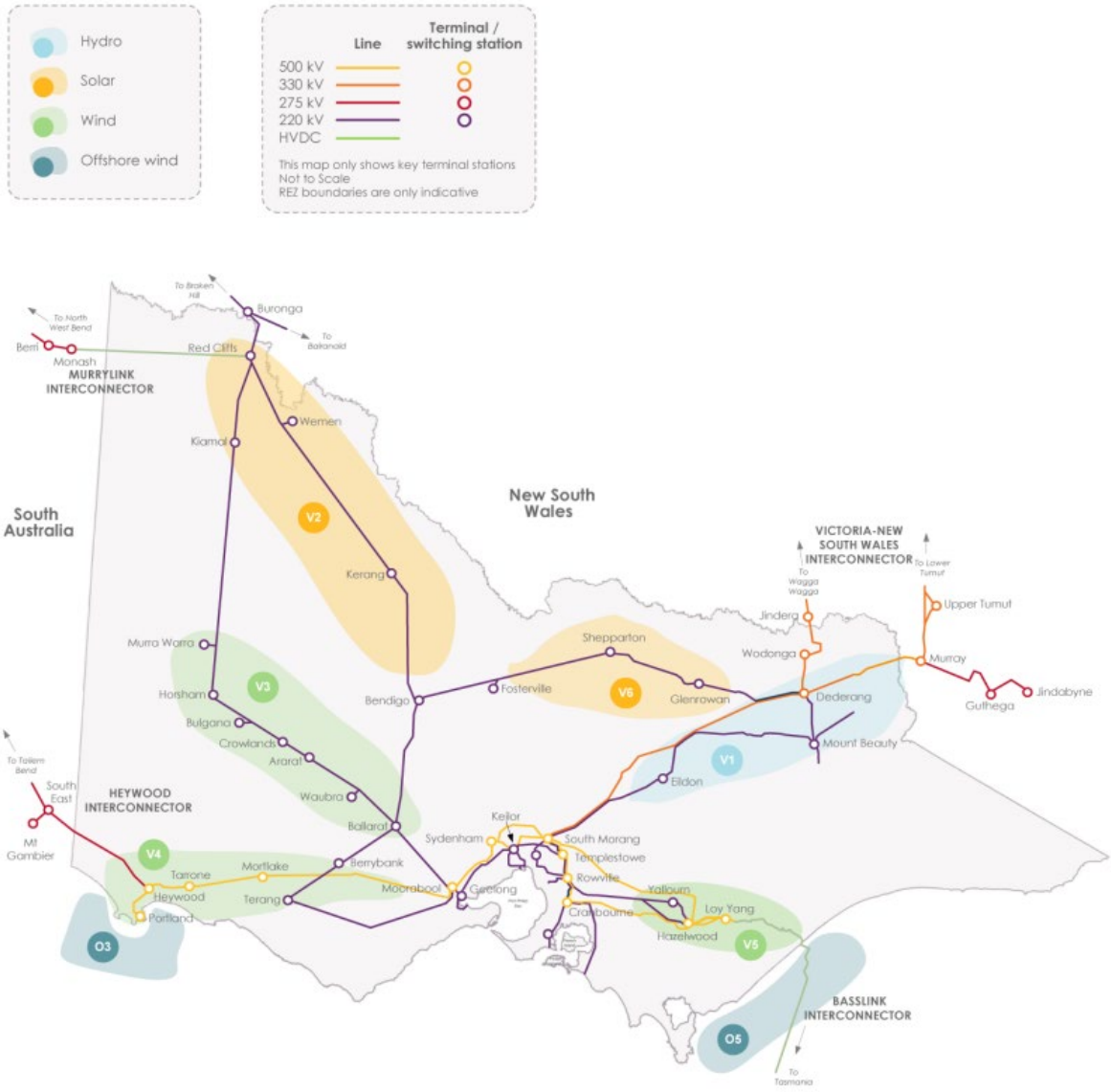
2.1. Electricity

The Victorian electricity system (**Figure 1**) is part of the National Electricity Market which also includes Queensland, New South Wales, Tasmania, South Australia, and the Australian Capital Territory. The National Electricity Market is operated by the Australian Energy Market Operator (AEMO). AEMO is responsible for reliability and system security consistent with its obligations under the National Electricity Law and the National Electricity Rules.

The Victorian electricity system includes:

- generators – coal and gas-fired power plants, and solar, wind and hydro generation
- energy storage – large-scale grid-connected batteries
- transformers, switching and protection equipment –voltage network control
- transmission lines – high voltage lines connecting the large generators to demand centres
- interconnectors – the transmission systems linking the Victorian network with the other National Electricity Market regions (Tas, SA, NSW)
- distribution networks – low voltage networks that connect supply to customers
- energy resources – behind the meter assets (solar panels, inverters, small capacity batteries)

Figure 1: Victorian Transmission Lines



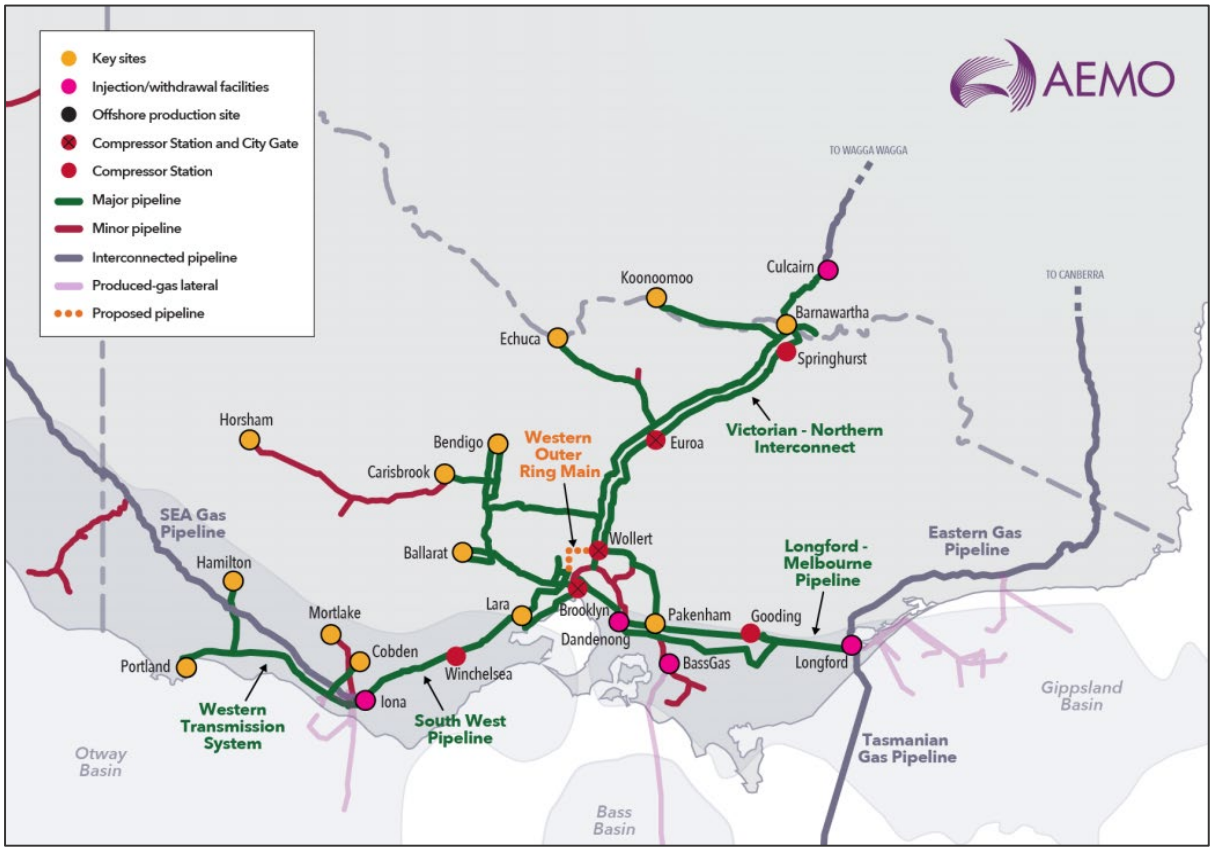
2.2. Gas

The Victorian gas system, most of which is within the Victorian Declared Wholesale Gas Market, forms part of the eastern gas region which is one of two regions in Australia's domestic gas market. The Victorian Declared Wholesale Gas Market enables dynamic trading of gas injections and withdrawals from Victoria's gas transmission system, known as the Declared Transmission System. AEMO operates both the Victorian Declared Wholesale Gas Market and the Declared Transmission System.

There are several major gas pipelines that fall outside of the (**Figure 2**), including the Eastern Gas Pipeline from Longford to Sydney, the SEAGas pipeline connecting South Australia to Victoria, the Tasmanian Gas Pipeline connecting Victoria and Tasmania and pipelines to some regional Victorian centres.

The broader gas supply system in Victoria consists of three gas basins off the coast (Gippsland, Bass and Otway basins), gas processing plants (Longford, Orbost, Port Campbell and Lang Lang), a small liquefied natural gas storage facility in Dandenong, the Iona underground natural gas storage facility in Port Campbell, and the transmission and distribution pipeline systems that deliver gas to customers.

Figure 2: Victorian Declared Gas Transmission System, Eastern, SEA and Tasmanian Gas Pipelines



(Image – AEMO)

2.3. Liquid Fuels

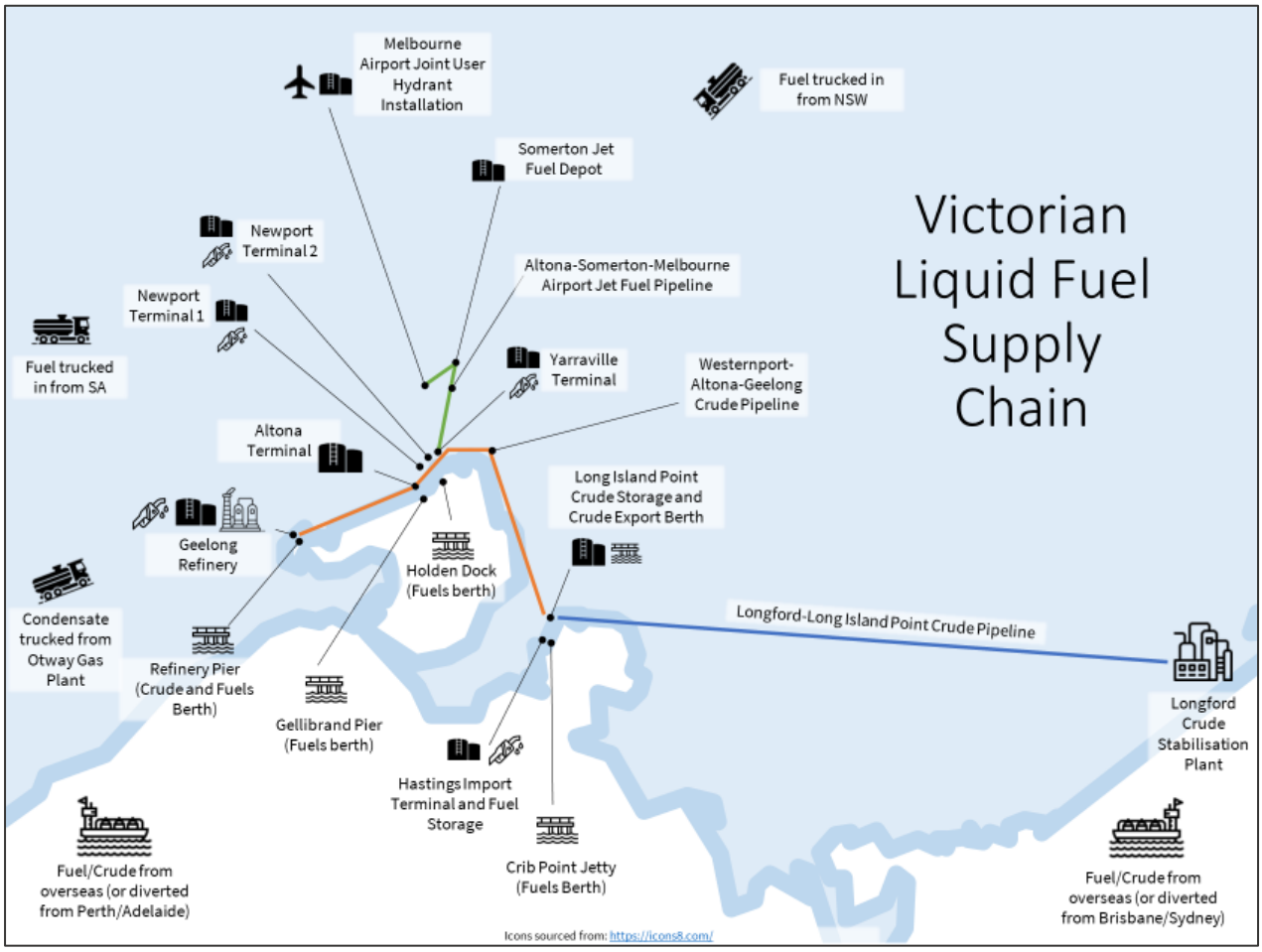
The liquid fuel sub-sector operates within a global supply chain through which Australia imports more than 90 per cent of its liquid fuel needs (both crude oil and refined fuel) from more than 60 countries.

The Victorian fuel supply chain (**Figure 3)** incorporates:

- offshore production of crude oil and condensate
- crude and condensate refining capacity

- import and export infrastructure for crude and refined product (e.g., berthing facilities and pipelines)
- transmission pipeline infrastructure including pipelines to Melbourne Airport
- storage and distribution infrastructure at the refinery in Geelong, terminals and regional fuel storage depots
- road tankers and logistics services
- a retail service station network.

Figure 3: Liquid Fuel Supply Chain[1]



[1] Icons sourced from: https://icons8.com/

3. Energy Risks and Dependencies

The responsibility for managing emergency risks is shared across government, industry, community and risk owners. Risk owners are any person or entity with the authority to manage a particular risk and who is accountable for doing so. The energy industry is subject to broad risk management requirements. Each distribution business identifies and manages corporate and emergency risks for their own operations and infrastructure. These are guided by their own risk management frameworks, which detail the assessment of risks, identification of existing and proposed controls and treatments where applicable.

Legislation at national and state level imposes obligations upon asset owners and/or operators of the energy system to demonstrate risk assessments and mitigate and prepare for emergencies.

Risk management planning for critical infrastructure includes consideration of physical facilities, supply chains, systems, assets, information technologies and communication networks, which if destroyed, degraded or rendered unavailable for an extended period, would significantly impact the Victorian community.

3.1. Government assessment and monitoring of risks

At a state level, Emergency Management Victoria (EMV) is responsible for coordinating a state-wide risk assessment that identifies the highest emergency risks and the controls and treatments that minimise or manage the risks and consequences. EMV's Emergency Risks in Victoria Report (2020) identifies electricity supply disruption and gas supply disruption as some of the highest risks in Victoria.

DEECA undertakes an emergency risk assessment every three (3) years and completes an annual review in the intervening years in accordance with the EM Act 2013 (Part 7A). The emergency risk assessment informs the development of the annual Energy Sector Resilience Plan and are incorporated into this sub-plan.

The risk assessment is undertaken in accordance with the principles of the National Emergency Risk Assessment Guidelines, which are consistent with international standard ISO31000:2018: Risk Management – Principles and Guidelines.

This risk assessment is informed by:

- advice from industry and other stakeholders
- consultation with the Sector Resilience Network - Energy convened by DEECA
- consultation with Energy Sector Group Trusted Information Sharing Network convened by the Commonwealth Government's Department of Home Affairs
- information included in 'vital' energy critical infrastructure businesses' Statements of Assurance
- review of the emergency risk environment and energy emergency events nationally and globally
- Victorian Government state level risk management priorities and risk assessments.

In 2022, the three yearly emergency risk assessment, assessed risks for each energy sub-sector (electricity, gas and liquid fuels) recognising that not all risks affect each sub-sector. In addition, vulnerabilities and consequences of each identified risk can be unique and discrete for electricity, gas and liquid fuels. As part of the sub-sector assessment, analysis confirmed upstream and downstream supply chain dependencies and the cascading affects major emergencies may have on these dependencies.

Risks and vulnerabilities identified in the emergency risk assessment informs this sub-plan and are provided in **Tables 2-4**.

3.2. Electricity

Table 2 provides the risks and consequences to electricity from natural hazard events, damage to critical electricity infrastructure and/or systems, disruptions to upstream or downstream dependencies,

supply chain disruptions, and interruptions to industry workforce. Extreme risks to electricity were identified as fire (bushfire/grassfire), extreme temperatures (heatwave) (3–4-day event), flood and major plant failure.

Electricity is critical for the operation of critical infrastructure across Victoria's eight (8) sectors and the provision of essential services to the State (downstream dependencies). In turn, electricity is dependent on the operation of the water, communications, and transport sectors to maintain supply of electricity upstream dependencies.

TABLE 2: Electricity Risk Ratings

Risk Rating	Risk Statement
Extreme	
Fire (including bushfire/grassfire)	There is the potential for bushfires to damage or destroy electricity infrastructure, or restrict electricity service operations within normal parameters, that, in turn, may cause an extended or expansive service delivery interruption.
Extreme temperatures (heatwave) (3–4-day event)	There is the potential that extreme temperatures will damage, or exceed the operating limit of, electricity infrastructure, that, in turn, may cause an extended or expansive service delivery interruption.
Floods	There is the potential for flood to damage electricity infrastructure, cause the loss of access/control, or restrict electricity service operations within normal parameters that, in turn, may cause an extended or expansive service delivery interruption.
Major plant failure – operations (including ageing infrastructure, maintenance, defects, operating outside of design)	There is the potential for a major plant failure (e.g. through asset management failure, design flaw, construction defect or out-of-design operation) to compromise electricity infrastructure integrity or restrict electricity service operations within normal parameters that, in turn, may cause an extended or expansive service delivery interruption.
High	
Severe weather (including extreme rain, lightning, hail and extreme wind)	There is the potential for a severe weather event to damage electricity infrastructure or restrict electricity service operations within normal parameters, that, in turn, may cause an extended or expansive service delivery interruption.
Cyber attack	There is the potential for a cyber-attack (inadvertent or malicious) to penetrate corporate IT, Supervisory control and data acquisition (SCADA) or asset-specific management systems and cause a loss of electricity service operational control that, in turn, may cause an extended or expansive service delivery interruption.
Coal Mine failure	There is potential for coal mine failure to compromise infrastructure integrity or restrict electricity service operations within normal parameters that, in turn, may cause an extended or expansive service delivery interruption.
Pandemic	There is the potential for a pandemic to lead to workforce constraints (e.g. via extended illness or isolation/quarantine) or compromise electricity service operational capabilities, that may in turn cause an extended or expansive service delivery interruption.
Grid instability	There is potential for grid instability to restrict the ability to maintain appropriate grid reliability conditions, (e.g. inertia), that, in turn, may cause an extended or expansive service delivery interruption.
Gas supply disruption	There is the potential for a gas supply disruption (e.g. damage to gas assets and infrastructure) or a loss of gas supply, to restrict electricity (gas-powered) generation or electricity service operations that, in turn, may cause an extended or expansive service delivery interruption.
Communications disruption	There is the potential for a communications disruption (e.g. damage to fibre optic, ASDL/NBN/cable, SCADA, telemetry) or a loss of a communications service (e.g. 3G/4G/5G) to impact electricity service remote operational monitoring and control, that, in turn, may cause an extended or expansive service delivery interruption.
Medium	

Risk Rating	Risk Statement
Earthquake	There is the potential for an earthquake to damage or destroy electricity infrastructure, or restrict electricity service operations within normal parameters, that, in turn, may cause an extended or expansive service delivery interruption.
Fire (structural)	There is the potential for a structural fire to damage or destroy electricity infrastructure, or restrict electricity service operations within normal parameters, that, in turn, may cause an extended or expansive service delivery interruption.
Physical intrusion or attack	There is the potential for a physical intrusion (criminal act by an internal or external person/s) by disruption, vandalism, terrorism, hostage situation or personal harm, to cause (or threaten to cause) a loss of operational control or restrict electricity service operations within normal parameters, that, in turn, may cause an extended or expansive service delivery interruption.
Unplanned closure of coal fired electricity generation	There is potential for an unplanned closure of coal fired power plants to cause a loss (in part or otherwise) of baseload energy capacity, that, in turn, may cause an extended or expansive service delivery interruption.
Foreign interference	There is the potential for foreign interference to compromise electricity service operational capabilities or capacity, that, in turn, may cause an extended or expansive service delivery interruption.
Workforce issues	There is the potential for workforce constraints (e.g. via unavailability due to fatigue/illness, major and prolonged industrial action, skills shortages, movement of workers or mass resignation/retirement) to impact electricity service operational capabilities and/or capacity, that, in turn, may cause an extended or expansive service delivery interruption.
Water supply disruption	There is the potential for a water supply disruption (e.g. damage to water assets and infrastructure) or a loss of water supply, to restrict electricity generation or service operations that, in turn, may cause an extended or expansive service delivery interruption.
Critical parts/spares disruption	There is the potential for a critical parts/spares disruption to compromise infrastructure integrity or restrict electricity service operations within normal parameters that, in turn, may cause an extended or expansive service delivery interruption.
Low	
Unexpected load / demand increase	There is the potential for an unexpected increase in load or demand to restrict the ability to maintain appropriate grid reliability conditions (e.g. inertia), that, in turn, may cause an extended service delivery interruption.
Liquid fuel disruption	There is the potential for a fuel shortage to result in reduced availability to gas-powered generators if compounded with a gas shortage or supply disruption that, in turn, may cause a supply demand imbalance during peak demand.
Transport network disruption	There is the potential for a transport network disruption to compromise supply chain/s and/or electricity service operation and maintenance access, that, in turn, may cause an extended service delivery interruption.
Solar flare	There is the potential for a solar flare/storm to damage or destroy electricity infrastructure, or restrict electricity service operations within normal parameters that, in turn, may cause an extended service delivery interruption.
Very Low	
Dam failure (hydroelectric power generation)	There is potential for dam failure to compromise infrastructure integrity or restrict electricity service operations within normal parameters that, in turn, may cause an extended or expansive service delivery interruption.

3.3. Gas

Gas in Victoria is primarily used for residential and commercial space heating and hot water, cooking and industrial process heating. If a gas supply disruption occurred in the summer, it could impact electricity peak demand for gas-powered electricity generation. Throughout the year, there is a potential impact to the wellbeing of people and all critical infrastructure sectors who rely on gas to heat homes and commercial premises.

The risks to gas availability (**Table 3**) are influenced by reporting undertaken by AEMO on supply and demand balances. In addition, risks are heightened from declining resilience in gas infrastructure from multiple factors, including aging infrastructure, existing pipeline constraints, reduced redundancy at the Longford gas plant, a decline in overall production of legacy fields and low levels of gas stored at Dandenong.

Major plant failure therefore was assessed as an extreme risk which is driven by an increase in the likelihood of failure under unlikely and likely scenarios from aging infrastructure and challenges with energy transition.

Gas is a vital upstream dependency for gas-powered generation and the start-up of thermal generators. Unplanned generator outages can significantly increase the demand for gas generation. Significant disruption to natural gas supply may require AEMO to direct gas-powered generation to shift to diesel generation or curtail gas-powered generation, increasing the risk of disruptions to the electricity supply.

TABLE 3: Gas Risk Ratings

Risk Rating	Risk Statement
Extreme	
Major plant failure - operations	There is the potential for a major plant failure (e.g. through asset management failure, design flaw, construction defect or out-of-design operation) to compromise gas infrastructure integrity or restrict gas service operations within normal parameters that, in turn, may cause an extended service delivery interruption.
High	
Fire (including bushfire, grassfire)	There is the potential for bushfires to damage or destroy gas infrastructure, or restrict gas service operations within normal parameters, that, in turn, may cause an extended or expansive service delivery interruption.
Extreme temperatures (cold) (3-4 day event)	There is the potential for extremely cold temperatures to increase gas demand over available supply capacity, and/or exceed the operating limit of gas infrastructure, that, in turn, may cause an extended or expansive service delivery interruption.
Cyber-attack	There is the potential for a cyber-attack (inadvertent or malicious) to penetrate corporate IT, SCADA or asset-specific management systems, cause a loss of gas service operational control that, in turn, may cause an extended or expansive service delivery interruption.
Fire (structural)	There is the potential for a structural fire to damage or destroy gas infrastructure, or restrict gas service operations within normal parameters, that, in turn, may cause an extended or expansive service delivery interruption.
Pandemic	There is the potential for a pandemic to lead to workforce constraints (e.g. via extended illness or isolation/quarantine) or compromise gas service operational capabilities and/or capacity, that, in turn, may cause an extended or expansive service delivery interruption.
Global conflict	There is potential for global conflict to reduce or restrict CAPEX or OPEX investment in gas infrastructure, that, in turn, may cause an extended or expansive service delivery interruption.
Medium	
Earthquake	There is the potential for an earthquake to damage or destroy gas infrastructure, or restrict gas service operations within normal parameters, that, in turn, may cause an extended or expansive service delivery interruption.

Risk Rating	Risk Statement
Extreme temperatures (heatwave) (3–4-day event)	There is the potential that extreme temperatures will damage, or exceed the operating limit of, gas infrastructure, that, in turn, may cause an extended or expansive service delivery interruption.
Severe weather (including extreme rain, lightning, hail, and extreme wind)	There is the potential for a severe weather event to damage gas infrastructure or restrict gas service operations within normal parameters, that, in turn, may cause an extended or expansive service delivery interruption.
Physical intrusion or attack	There is the potential for a physical intrusion (criminal act by an internal or external person/s) by disruption, vandalism, terrorism, hostage situation or personal harm, to cause (or threaten to cause) a loss of operational control or restrict gas service operations within normal parameters, that, in turn, may cause an extended service delivery interruption.
Workforce issues	There is the potential for workforce constraints (e.g. via unavailability due to fatigue/illness, major and prolonged industrial action, skills shortages, movement of workers or mass resignation/retirement) to impact gas service operational capabilities and/or capacity, that, in turn, may cause an extended or expansive service delivery interruption.
Electricity disruption	There is the potential for an electricity disruption (e.g. damage to electricity assets and infrastructure) or a loss of electricity supply to restrict gas service operations that, in turn, may cause an extended or expansive service delivery interruption.
Communications disruption	There is the potential for a communications disruption (e.g. damage to fibre optic, ASDL/NBN/cable, SCADA, telemetry) or a loss of a communications service (e.g. 3G/4G/5G) to impact gas service remote operational monitoring and control, that, in turn, may cause an extended or expansive service delivery interruption.
Critical parts/spares disruption	There is the potential for a critical parts/spares disruption to compromise infrastructure integrity or restrict gas service operations within normal parameters that, in turn, may cause an extended or expansive service delivery interruption.
Low	
Floods	There is the potential for flood to damage gas infrastructure, cause the loss of access/control, or restrict gas service operations within normal parameters that, in turn, may cause an extended or expansive service delivery interruption.
Liquid fuel disruption	There is the potential for a liquid fuel disruption to restrict personnel mobility and cause a loss of operational control of gas service infrastructure (during an electrical outage), that, in turn, may cause an extended service delivery interruption.
Transport network disruption	There is the potential for a transport network disruption to compromise supply chain/s and/or gas service operation and maintenance access, that, in turn, may cause an extended service delivery interruption.

3.4. Liquid Fuels

The overall risk to liquid fuel supplies in Victoria (**Table 4**) are not considered as significant as the risks to electricity and gas supply, because it is unlikely an event would lead to major disruption to service continuity. Despite this, the risk profile in the sector remains dynamic and subject to change over time. The risks to liquid fuel supply that have been assessed as high included cyber-attack, pandemic, and transport network disruption, all of which could impact service continuity if materialised.

Liquid fuel supplies are critical for electricity and gas generators, including back up for gas-powered generation and for elements of coal-mining operations, which coal-fired power stations depend on. This is particularly the case in the event of a gas shortage where some gas-powered generation may switch to diesel in response to market conditions.

The transport, agriculture and mining sectors more broadly are also highly dependent on liquid fuel for operations, maintenance, repairs and supply by road, rail and sea. Any major disruptions to liquid fuel supply are likely to have cascading impacts across multiple critical infrastructure sectors.

TABLE 4: Liquid Fuels Risk Ratings

Risk Rating	Risk Statement
Extreme	
Nil	
High	
Cyber attack	There is the potential for a cyber attack (inadvertent or malicious) to penetrate corporate IT, SCADA or asset-specific management systems, cause a loss of liquid fuel service operational control that, in turn, may cause an extended or expansive service delivery interruption.
Pandemic	There is the potential for a pandemic to lead to workforce constraints (e.g. via extended illness or isolation/quarantine) or compromise liquid fuel service operational capabilities and/or capacity, that, in turn, may cause an extended or expansive service delivery interruption.
Transport network disruption (ports maritime routes / road network)	There is the potential for a transport network disruption to compromise supply chain/s and/or liquid fuel service operation and maintenance access, that, in turn, may cause an extended service delivery interruption.
Medium	
Fire (including bushfire/grassfire)	There is the potential for bushfires to damage or destroy liquid fuel infrastructure, or restrict liquid fuel service operations within normal parameters, that, in turn, may cause an extended or expansive service delivery interruption.
Fire (structural)	There is the potential for a structural fire to damage or destroy liquid fuel infrastructure, or restrict liquid fuel service operations within normal parameters, that, in turn, may cause an extended or expansive service delivery interruption.
Physical intrusion or attack	There is the potential for a physical intrusion (criminal act by an internal or external person/s) by disruption, vandalism, terrorism, hostage situation or personal harm, to cause (or threaten to cause) a loss of operational control or restrict liquid fuel service operations within normal parameters, that, in turn, may cause an extended service delivery interruption.
Major plant failure – operations	There is the potential for a major plant failure (e.g. through asset management failure, design flaw, construction defect or out-of-design operation) to compromise liquid fuel infrastructure integrity or restrict liquid fuel service operations within normal parameters that, in turn, may cause an extended service delivery interruption.
Workforce issues	There is the potential for workforce constraints (e.g. via unavailability due to fatigue/illness, major and prolonged industrial action, skills shortages, movement of workers or mass resignation/retirement) to impact liquid fuel service operational capabilities and/or capacity, that, in turn, may cause an extended or expansive service delivery interruption.
Electricity supply disruption	There is the potential for an electricity disruption (e.g. damage to electricity assets and infrastructure) or a loss of electricity supply to restrict liquid fuel service operations that, in turn, may cause an extended or expansive service delivery interruption.
Communications disruption	There is the potential for a communications disruption (e.g. damage to fibre optic, ASDL/NBN/cable, SCADA, telemetry) or a loss of a communications service (e.g. 3G/4G/5G) to impact liquid fuel service remote operational monitoring and control, that, in turn, may cause an extended or expansive service delivery interruption.
Global conflict	There is potential for global conflict to reduce or restrict CAPEX or OPEX investment in liquid fuel infrastructure, that, in turn, may cause an extended or expansive service delivery interruption.
Critical parts/spares disruption	There is the potential for a critical parts/spares disruption to compromise infrastructure integrity or restrict liquid fuel service operations within normal parameters that, in turn, may cause an extended or expansive service delivery interruption.

OFFICIAL

Risk Rating	Risk Statement
Low	
Extreme temperatures (heatwave) (3-4 days)	There is the potential that extreme temperatures will damage, or exceed the operating limit of, liquid fuel infrastructure, that, in turn, may cause a service delivery interruption.
Floods	There is the potential for flood to damage liquid fuel infrastructure, cause the loss of access/control, or restrict liquid fuel service operations within normal parameters that, in turn, may cause an extended or expansive service delivery interruption.
Severe weather (including extreme rain, lightning, hail, and extreme wind)	There is the potential for a severe weather event to damage liquid fuel infrastructure or restrict liquid fuel service operations within normal parameters, that, in turn, may cause an extended or expansive service delivery interruption.
Earthquake	There is the potential for an earthquake to damage or destroy liquid fuel infrastructure, or restrict liquid fuel service operations within normal parameters, that, in turn, may cause an extended or expansive service delivery interruption.
Gas supply disruption	There is the potential for a gas supply disruption (e.g. damage to gas assets and infrastructure) or a loss of gas supply, to restrict liquid fuel service operations that, in turn, may cause a service delivery interruption.
Water supply disruption	There is the potential for a water supply disruption (e.g. damage to water assets and infrastructure) or a loss of water supply, to restrict liquid fuel production or service operations that, in turn, may cause an extended or expansive service delivery interruption.

4. Energy Emergencies

An energy emergency is any disruption of electricity, gas or liquid fuel supply that has an impact on the community, health system, economy, and/or the environment.

Disruptions can occur at any point in the energy supply chain. Localised and limited incidents impacting energy infrastructure and/or supplies are managed by industry and relevant response agencies as part of business-as-usual operations.

4.1. Major Emergencies

An **emergency,** as defined in the EM Act 2013 (s. 3), means an actual or imminent occurrence of an event which in any way endangers or threatens to endanger the safety or health of any person in Victoria or which destroys or damages, or threatens to destroy or damage, any property, the environment.

A '**major emergency**' is a large or complex emergency, with the potential to:

- cause loss of life, or extensive damage to property, infrastructure, or the environment; or
- have or is having significant adverse consequences for the Victorian community, or part of the Victorian community; or
- an emergency that involves two or more agencies to respond.

A 'Class 2' emergency means an emergency which is not a Class 1 or Class 3 emergency. Major energy emergency is the disruption to electricity, gas or liquid fuel supply and which meets the criteria of a 'major emergency' as defined by the EM Act 2013 (above). A proclamation or declaration does not need to be made for an emergency to be a Class 2 major emergency.

A major energy emergency may include one or more of the following:

- a large and complex incident that has the potential to have or is having significant adverse consequences to the safety and wellbeing of the Victorian community or a part of the community
- immediate threat to significant critical infrastructure or facilities, including hospitals, major industry and/or other utilities
- generation, transmission or distribution equipment failure or disruption of energy supplies to Victoria
- demand for electricity or gas significantly exceeds supply and requires market intervention by AEMO
- an event has occurred which may endanger an undertaking of a distribution business, a transmission business or a generation business or person or a person who supplies electricity, gas or liquid fuel
- substantial imbalances in the markets affecting primary grade fuel(s), such as diesel, unleaded petrol and/or jet fuel, requiring prioritisation for essential fuel users
- disruptions to supply chains that impact upon energy supplies in Victoria

4.2. Levels of energy emergencies

For localised[2] and limited[3] incidents, the relevant owner/operator is responsible for the management of the incident. This can include minor infrastructure damage or localised retail impact. DEECA monitors and provides subject matter expertise to emergency management agencies where required.

[2] **Localised** (Level 1) is defined as those incidents, likely or actual, that impact at a local level, are of short duration that can be resolved by local/initial response resources and does not impact upon significant critical infrastructure or facilities, including hospitals, major industry and other utilities

[3] **Limited** (Level 2) is defined as those incidents, likely or actual, that impact beyond a local area, that may include more than one distribution business or region and has the potential to affect significant infrastructure or facilities requiring incident management functions to be established

OFFICIAL

Table 5 summarises triggers and actions for control of localised, limited, and major impact energy emergencies that may apply to electricity, gas or liquid fuel incidents. Escalation of incidents from localised and limited levels to a major energy emergency is determined by forecasts[4] and how an incident[5] has or may have an impact upon critical infrastructure and provision of electricity, gas or liquid fuel supplies.

TABLE 5: Summary of triggers, actions, and control status

		Trigger	Action		Control
LOCALISED	Forecast	Weather forecast, market conditions or other risk assessment suggests likelihood of a localised event occurring	AEMO; relevant industry owners or operators monitor risk according to relevant plans	DEECA monitors; provides subject matter expertise as required.	Industry incident operations
	Actual	Disruption is localised, small scale and/or very short duration.	AEMO, relevant industry owner or operator manages, responds	DEECA monitors; provides subject matter expertise as required	Industry incident operations
LIMITED	Forecast	Weather forecast, market conditions or other risk assessment suggests likelihood of a limited impact emergency occurring	AEMO; relevant industry owners or operators monitor risk according to relevant plans	DEECA monitors; provides subject matter expertise as required	Industry incident operations
	Actual	Disruption has a limited geographic, economic, environmental or population impact and/or short duration.	AEMO and industry owners or operators manage response and may activate an IMT and provide intelligence to DEECA	DEECA monitors; provides subject matter expertise as required; resources available if event escalates	Industry incident operations
MAJOR	Forecast	Weather forecast, market conditions or other risk assessment suggests a likelihood of a major emergency event occurring	AEMO and industry owners/operators monitor Industry embed support at state, regional and/or incident tier.	SCC-Energy Intelligence Unit activated (readiness)	DEECA is control agency; State Controller – Energy may be appointed
	Actual	An incident has occurred that is widespread or complex with the potential to cause loss of life, or extensive damage to property, infrastructure, or the environment, or have significant adverse impact on the community, or part of the community; or require the involvement of two or more agencies to manage the emergency.	AEMO and industry owners/operators manage, respond. Industry embed support at state, regional and/or incident tier.	SCC-Energy Intelligence Unit activated (response and relief)	DEECA is control agency; State Controller – Energy may be appointed

[4] **Forecast** conditions combine weather forecasting and supply and demand predictions to determine if conditions will lead to lack of reserve for electricity and gas shortfall prompting AEMO to issue notices to the market to increase supply

[5] **Incidents** are conditions that are or likely to occur and triggers a response from AEMO, owners/operators and/or DEECA to respond to conditions that have or will result in a major energy emergency

5. Emergency Management Planning

The EM Act 2013 provides the context and planning footprint to all tiers of government with specific considerations that apply to each tier. The EM Act 2013 stipulates the following objectives to be built into emergency management planning:

- Foster a sustainable and efficient emergency management system that minimises the likelihood, effect and consequences of emergencies; and
- Establish efficient governance arrangements that –
    - Clarify the roles and responsibilities of agencies; and
    - Facilitate cooperation between agencies; and
    - Ensure the coordination of emergency management reform within the emergency management sector; and
- Implement an 'all communities-all emergencies" approach to emergency management; and
- Promote community resilience; and
- Establish integrated arrangements for emergency management planning in Victoria.

Further, the EM Act 2013 provides clauses of what emergency management plans must include.

State Emergency Management Planning

The Emergency Management Commissioner is responsible for the preparation of the SEMP.

This sub-plan forms part of the comprehensive arrangements for emergency management in Victoria and is a sub-plan to the SEMP.

Regional Emergency Management Planning

Regional Emergency Management Planning Committees are responsible for preparing the regional emergency management plan.

It is good practice that Regional Emergency Management Planning Committees invite subject matter experts, such as energy businesses, AEMO, DEECA, and Energy Safe Victoria to inform regional level planning. AEMO communicates with DEECA and industry, with information flow to regional centres via DEECA or participation in the Regional Emergency Management Planning Committees.

Municipal Emergency Management Planning

Municipal Emergency Management Planning Committees are responsible for preparing Municipal Emergency Management Plans. It is good practice that Municipal Emergency Management Planning Committees invite energy subject matter experts to inform municipal planning.

Critical Infrastructure Planning

The EM Act 2013 (Part 7A) states that the Minister for Energy and Resources is responsible for assessing the criticality of energy critical infrastructure. These assessments designate the level of criticality as 'vital' (highest), 'major', 'significant' or 'local'. The owners and operators of designated vital critical infrastructure are subject to additional requirements under Part 7A within the Resilience Improvement Cycle: emergency risk management planning, exercising, audits and assurance reporting.

Each year, owners/operators of designated vital critical infrastructure are required to undertake a Risk Management Plan. This plan provides those actions and activities each entity will undertake to ensure the mitigation of risk, preparedness for, response to and recovery from emergency events identified to ensure continued safe operation.

Sector Resilience Plans

Each year, as required under the EM Act 2013 (Part 7A), Sector Resilience Plans are prepared by each of the eight Sector Resilience Networks.

DEECA, as the government lead for the energy sector, works with industry to produce an Energy Sector Resilience Plan that includes:

- emergency risks per energy sub-type
- the potential consequences of those risks, and
- identifies resilience improvement initiatives for the following 12 months.

Sector Resilience Plans for other critical infrastructure sectors (water, food and grocery supply logistics, health, transport, communications, banking and finance, and government) consistently identify energy disruption as a key risk requiring preparedness and mitigation strategies at the sector-wide and individual operator levels.

Statement of Opportunities

Each year AEMO produces the Electricity Statement of Opportunities and Gas Statement of Opportunities. The Electricity Statement of Opportunities and the Gas Statement of Opportunities forecasts long-term reliability of electricity and gas supply to inform decisions by market participants, investors, and policy makers. By forecasting any potential reliability gaps over a forward outlook period, the Electricity Statement of Opportunities and the Gas Statement of Opportunities assist in supporting the planning to mitigate future supply risks.

Community Emergency Planning

The government and emergency services encourage all individuals and households to become aware of and prepare for any emergency that may disrupt supplies of electricity, gas or liquid fuels. To do so, a well-practiced written emergency plan is recommended which is informed by safety awareness campaigns, programs led by emergency services and community planning. **Table 6** provides details of a range of advice that is available to households and communities to assist in their emergency management planning.

TABLE 6: Resources for Households to Plan for Emergencies

Details of Information available	Links to available resources
Fire services encourage households to have a Bushfire Survival Plan, so residents know what to do when they are at risk of fire. These written plans should be well practiced so everyone in the household remembers what to do during a crisis. Other resources are available to assist households on what to do before and after a fire.	Bushfire Survival Plan Local Information and advice Resources to understand fire risk and what to do before and during a fire How to prepare your property
The Victorian State Emergency Service support community preparedness for floods and storms. Resources are provided to assist households develop a flood and storm plan, so residents know what to do before, during and after a flood or storm emergency.	Flood – Plan and Stay Safe Storms – Plan and Stay Safe Local Flood Guides Community Safety Program for Flood
The Department of Health provide a suite of communication resources to assist individuals and the community to be aware of the impact of extreme heat on human health	Extreme Heat – Community Resources
DEECA and electricity businesses encourage households to include measures on what to do during power outages, which can be built into your emergency plan for fire, flood and storm.	Power outages

6. Mitigation

Mitigation is the elimination or reduction of the incidence or severity of emergencies and the minimisation of their effects, and includes measures undertaken as part of business-as-usual functions to mitigate emergencies.

6.1. Roles and Responsibilities

Agencies with roles and responsibilities for mitigating disruptions to electricity, gas or liquid fuel supplies are set out in the SEMP Roles and Responsibilities.

The energy industry has a key role to mitigate energy emergencies. AMEO and owners and operators of infrastructure and supply chains undertake a range of activities to mitigate the likelihood and severity of energy disruption. This includes:

- Infrastructure design (including redundancy), maintenance, safe operation and physical security
- Supply and demand management, modelling, planning and strategies, and market mechanisms
- Industry forums, including regular emergency exercises

Energy owners and operators undertake their role in mitigation based on a range of legislation, regulations, policies and programs with government agencies performing a monitoring and assurance role.

All energy users from households, to business, industry and government departments have a role in mitigating the consequences from energy disruptions.

Government, emergency management agencies and other stakeholders provide advice to Victorians through a range of programs, information and tools on how to mitigate the risk of bushfires, storms, flood and heatwaves and consequences from disruptions to energy supplies.

Table 7 provides details of mitigation actions for which government, owners/operators of critical infrastructure, and the Victorian community are doing to mitigate the consequences of energy emergencies.

TABLE 7: Mitigation measures to risks

Approach	Key actions
Reduce incidents of power outages from vegetation	Education and advertising campaigns (clearing vegetation close to powerlines, monitoring height and health of trees near powerlines)
	Coordinated, vegetation clearance (both on individual properties, easements and roads)
	Deterrence for non-compliance of vegetation clearance within legislative obligations
	Installation, operation, and maintenance of Rapid Earth Fault Current Limiters to reduce bushfire risk from powerlines
	Undergrounding of powerlines in high-risk bushfire areas
	Maintenance and/or undergrounding of privately owned overhead powerlines
Reduce the impact and consequences of storm damage that have led to power outages	Emergency plan and kit (reducing incidents from storm events on powerlines, alternative power supplies, plans for power dependent customers)
	Education and advertising campaigns (limiting use of electricity, alternative ways to keep cool, preparing for power outages)
Prepare for the impacts of extreme temperatures (heatwave and cold)	Emergency plan and kit (alternative ways to keep cool, plans to relocate, alternative power supplies for power outages, limiting use of electricity)
	Pre-planning for demand and supply from overuse of electricity (Lack of Reserve forecasting, load shedding strategies, voluntary gas curtailment)

OFFICIAL

Approach	Key actions
	Education and advertising campaigns (e.g. preparing properties for bushfire)
Prepare for the impacts of bushfire	Emergency plan and kit (plans to relocate, alternative power supplies, limiting use of electricity)
	Pre-planning for reducing ignition of bushfires from powerlines (e.g. Total Fire Ban Days, Rapid Earth Fault Current Limiters operations)
	Pre-planning for the impacts of bushfire on electricity (prepositioning of large-scale batteries, preposition of generators to maintain telecommunications, water supplies, relief centres, alternative power supplies)
	Vegetation management on public and private land in the vicinity of powerlines
Reduce the duration of power outages	Access and egress of arterials, roads, tracks and easements for repairs to electricity infrastructure
	Education and advertising campaigns (alternative power supplies, plans to relocate, plans for vulnerable persons, safety risks of using generators and outdoor heaters inside)
Reduce the health effects of prolonged power outages and gas disruptions on communities	Emergency plan and kit (plans to relocate, alternative power supplies, limiting use of electricity)
	Recovery planning and relationship building pre-prolonged power outages (via community groups, scenario events and activities)



Credit Energy Safe Victoria

7. Preparedness

Preparedness are those activities that government, the energy industry and community can do to prepare for and reduce the effects of emergencies. Services, businesses (of all sizes) and critical infrastructure with a dependency on electricity, gas and liquid fuels (such as manufacturing, hospitals, water corporations, transportation and other critical infrastructure) need to ensure they prepare and regularly test business continuity plans to reduce the risk of energy disruption.

7.1. National Energy Objectives

The Australian Energy Market Commission is an independent statutory body that develops and maintains the national energy legislation. These laws impact upon how energy market participants operate in the competitive generation and retail sectors, provide specific protections for consumers, and govern economic regulation of electricity transmission and distribution services and gas pipelines.

The National Electricity Objective promotes efficient investment in, efficient operation and use of, electricity services for the long-term interests of consumers of electricity with respect to:

- price, quality, safety, and reliability[6] and security of supply of electricity
- the reliability, safety, and security[7] of the national electricity system.

The National Gas Objective promotes efficient investment in, and efficient operation and use of, natural gas services for the long-term interests of consumers of natural gas with respect to price, quality, safety, reliability and security of supply of natural gas.

7.2. Guaranteed Service Levels

The Australian Energy Regulator administers the Service Target Performance Incentive Scheme that applies to electricity distribution networks. The Service Target Performance Incentive Scheme provides these networks with incentives for maintaining and improving network performance and achieve better reliability outcomes for all customers, including those in regional and rural areas.

In Victoria, the Essential Services Commission administers the Electricity Distribution Code of Practice including the Guaranteed Service Level scheme. Under the Guaranteed Service Level, customers who experience many or long unplanned power outages during a year may be entitled to automatic annual compensation payments.

7.3. Electricity and Gas Safety

Energy Safe Victoria (ESV) is the independent safety regulator responsible for electricity, gas and pipeline safety in Victoria. ESV's role is broad and ranges from accepting industry's safety cases and safety management schemes for the design, construction and maintenance of electricity, gas and pipeline networks. ESV performs a regulatory role against standards and administers regulations covering gas and electrical appliances and installations.

ESV regulates the legislative requirements that electricity distribution businesses develop, submits, and comply with an Electricity Safety Management Scheme, five yearly Bushfire Mitigation Plans and an annual electric line clearance management plan.

In addition, ESV conducts comprehensive public awareness safety campaigns to mitigate the risks to the community from gas and electricity emergencies.

Gas and liquid fuel pipelines are regulated under the *Gas Safety Act 1997* and *Pipelines Act 2005*. ESV and DEECA exercise regulatory powers and oversight under this legislation and associated regulations.

[6] Reliability is associated with consumer expectations regarding the dependability of their supply.

[7] Security is part of the technical resilience of the system.

Under the Safety Case and Safety Management Planning regime, gas companies have an integrated governance structure with risk and asset management processes ensuring the integrity, safe operations and reliability of natural gas system from hazards and risks.



Credit Energy Safe Victoria

7.4. Major Hazard Facilities

Owners and/or operators of the refinery, fuel terminals and gas terminals are required to prepare a safety case to obtain and maintain a licence to operate a major hazard facility. A major hazard facility is an industrial site that stores, handles or processes large quantities of chemicals and dangerous goods that are essential to the state's infrastructure through the provision of products and services.

Under Part 5.2 of the *Occupational Health and Safety Regulations (2017)* major hazard facility operators must identify and prevent or control major incidents that could occur, ensuring protection for:

- workers at the facility
- the surrounding community
- the infrastructure that the facility needs to keep operating safely.

In accordance with the Occupational Health and Safety Regulations, major hazard facility operators are required to gain a licence which responds to a safety assessment to all hazards identified.

Under the *Environment Protection Act 2017*, major hazard facilities must undertake a risk assessment of any hazards that are likely to be a risk of harm to human health and the environment from pollution or waste. In accordance with the *Environment Protection Act 2017* owners and operators of major hazard facilities are required to comply with and meet the General Environmental Duty.

7.5. Preparedness for weather conditions and natural hazard events

In addition to the legislative and regulatory requirements outlined above, arrangements are in place on how DEECA, AEMO, owners/operators of critical infrastructure prepare for weather conditions or potential or actual natural hazard events. **Table 8** provides specific preparedness arrangements for

government, emergency services and other stakeholders to apply in the lead up to conditions that may result in a major emergency and how these arrangements are implemented by the energy sector.

TABLE 8: Preparedness Arrangements

	Emergency Management Arrangements	Application to Energy Sector
Total Fire Ban (TFB)	• Total Fire Ban(s) are declared by the CFA Chief Officer under the CFA Act 1958 within one or more fire districts • Determined by forecasted extreme temperatures, humidity and wind direction/speed • Increases the likelihood of fires occurring with increased risk and consequences • Declaration imposes restrictions prohibiting fires to be lit in the open air	• AEMO monitors electricity supply and demand to mitigate the risk of lack of reserve • Electricity planned maintenance re-scheduled • Defective privately owned power lines deenergised • Increase in sensitive of Rapid Earth Fault Current Limiters to prevent fire ignition from powerlines
Catastrophic Day(s)[8]	• Determined by the interaction of fuel, weather and topography determining bushfire behaviour • The BOM uses weather variables to calculate indices for both grassland and forest fuels to estimate fire behaviour • Restrictions imposed on Total Fire Ban(s) apply on Catastrophic Days • May be declared for one or more fire districts • A Catastrophic Days may be declared within a district from a point in time if conditions change	• DEECA liaises with AEMO, ESV and the energy sector in the lead up to and during a Catastrophic Day • Transmission and distribution businesses across the electricity, gas and liquid fuel sectors enact Catastrophic Day arrangements • DEECA plans to activate the Victorian Electricity Emergency Communications Protocol in readiness for a Catastrophic Days • An energy industry liaison may be deployed to the SCC upon request of DEECA
Extreme Temperature (Heatwaves)	• Extreme heat is a sustained period of high temperature (heatwave) or a single day of higher than average temperature for that time of the year • Heatwaves are three days or more of high maximum and minimum temperatures that are unusual for a given location • The effect of heat is cumulative on the community, infrastructure, and services • The Bureau of Meteorology issue Heatwave Warnings, while Department of Health may issue a Heat Health Alert based on temperature thresholds	• DEECA liaises with AEMO, ESV and the energy sector in the lead up to and during a heatwave • Transmission and distribution businesses enact Heatwave/Total Fire Bang arrangements • An energy industry liaison may be on standby in readiness for deployment to the State Control Centre upon request of DEECA
Fire, Storm, & Flood	• May coincide or follow extreme weather conditions • Direct impact may result on critical infrastructure assets, disruptions to supply	• DEECA liaises with AEMO and the energy sector in the lead up to and during forecasted severe weather events • An energy industry liaison may be on standby in readiness for deployment to the State Control Centre upon request of DEECA • Mutual Aid Plans for electricity distribution businesses may be enacted to increase capacity to respond to power outages

[8] A catastrophic fire danger rating means conditions are the most dangerous for fire and advises the public to leave bushfire risk areas

8. Readiness

Readiness increases the ability for timely response with activities immediately before and in anticipation of a potential or actual emergency. Command, control, and coordination arrangements must be in place to ensure readiness:

- for major emergencies irrespective of whether a specific emergency is anticipated, occurring, or has occurred
- if a major emergency is anticipated, such as forecasts that indicate the possibility of a class 1 hazard or class 2 energy emergency
- if a limited and localised incidents that could potentially escalate into a major emergency.

8.1. Roles and Responsibilities

There are well established National and State emergency management arrangements and protocols in place in the energy sector. These arrangements link DEECA, AEMO, and energy owners/operators for the monitoring of risks of energy disruption through a range of mechanisms. Activation of DEECA's readiness arrangements is based on:

- forecasting of supply and demand conditions for the seven (7) day outlook
- monitoring of weather forecasts for heatwave, high fire danger, damaging winds and storm activity
- monitoring of any major emergency that has or is likely to have consequences on energy infrastructure, systems, or supplies

8.1.1 Electricity

AEMO produces forecasts for the day-to-day operations of the National Electricity Market covering forecasts for eight days ahead. These include forecasts for intermittent renewable generation for wind and solar, as well as electricity demand forecasts. Operational forecasts are critical inputs into the projected assessment of system adequacy and essential for maintaining power system security.

AEMO undertakes year-round, 24-hour operations to manage the National Electricity Market and are constantly monitoring the system to ensure security of supply and system strength. In addition, electricity distribution businesses are constantly monitoring their assets to ensure safe and effective operations.

8.1.2 Gas

AEMO provides planning and forecasting information to support efficient decision-making and long-term investment in Australia's gas markets and infrastructure services. AEMO publish Victorian Gas Planning Report which informs market participants and government about gas transmission capacity changes that impact security and reliability in the Victorian Gas Declared Transmission System.

AEMO undertakes year-round, 24-hour operations of the Declared Transmission System to ensure its safe and effective operation, and schedules the Victorian Declared Wholesale Gas Market. Gas pipeline operators, storage facilities, distributors, and producers are constantly monitoring equipment and systems to ensure safe and effective operations.

8.1.3 Liquid Fuels

Stocks and usage of liquid fuels are monitored by liquid fuel companies who report regularly to the Commonwealth Government in line with mandatory reporting requirements. The Commonwealth Government collates and publishes these statistics monthly, which can be useful for identifying long-term trends and emerging risks related to stock levels. Liquid fuel asset owners and operators also monitor their systems as part of ongoing operations and safety protocols.

The Commonwealth Government has also introduced a Minimum Stockholding Obligation, to commence on 1 July 2023, requiring major fuel importers and refineries to hold baseline stocks of petrol, diesel and jet fuel.

Further owners and operators of vital critical liquid fuel infrastructure in Victoria must comply with Part 7A of the EM Act 2013, including but not limited to maintaining an emergency risk management plan and conducting annual emergency exercises.

8.2. Readiness for weather conditions and natural hazard events

Readiness arrangements for the energy industry may be activated as a singular energy emergency or in response to forecast conditions or may coincide with a class 1 major emergency that has cascading effects on the safe and reliable supply of electricity, gas or liquid fuels. Therefore, in addition a range of measures are implemented in readiness for a Class 1 emergency that may have an impact on energy supplies or critical infrastructure.

Table 9 details those arrangements of government, emergency services and other stakeholders to apply in readiness for a potential or actual major emergency (Class 1) and what arrangements are implemented by DEECA, AEMO, and energy owners/operators.

TABLE 9: Readiness Arrangements

	Emergency Management Arrangements	Application to Energy Sector
Fire in the landscape **Severe weather** **Flood** **Heatwave** **Earthquake**	• VICSES, CFA, FRV or FFMVic Controller • State Control Centre activated at Tier 2 or 3 • Incident Control Centre or Regional Control Centre may request Energy-Emergency Management Liaison Officer • DEECA State Agency Commander activates State Control Centre -Energy Unit based on potential or actual emergency triggers being met or at the request of State Response Controller • State Response Controller may appoint a Deputy Controller - Energy	• State Control Centre -Energy Unit activated • State Control Centre -Energy Unit liaise with Fire Behavioural Analyst or Flood Analyst to monitor risks to critical energy infrastructure • State Control Centre -Energy Unit monitor weather forecasts for further risks • Critical infrastructure sectors request priority site restoration of electricity • DEECA Secretary as Control Agency Officer in Charge may appoint State Controller – Energy • State Control Centre -Energy Unit liaise with AEMO, distribution and transmission businesses on risks and potential consequences



Image Iona Gas Storage Facility. Credit Lochard Energy

Box 1 demonstrates an example of ongoing risk management and readiness in respect to Victoria's gas storage reserves and contingencies implemented to ensure there are no shortfalls of gas and what conditions contribute to a major gas emergency.

Box 1: IONA Underground Storage Case Study

The Iona Underground Gas Storage (Iona) facility provides critical support in meeting peak winter gas demand in Victoria.

During winter 2022, there were multiple coal power station outages and low wind power generation across Queensland, NSW and Victoria. At the same time, Victoria experienced periods of cold weather. Gas demand from gas powered electricity generation increased and the Iona storage experienced above average withdrawals.

Above average withdrawals in May depleted Iona storage from a maximum storage of 23,599 TJ on 18 April 2022 to 9,493 TJ on 20 July 2022, or 40 per cent of total capacity. These levels were lower than at any point the previous year and approaching a previous record low storage of 8,565 TJ on 7 September 2018.

As the storage level decreases, factors relating to physical characteristics of the storage facility result in a reduction of the quantity of gas that can be injected into the Declared Wholesale Gas Market (DWGM). Under normal operating conditions Iona can inject up to 530 TJ/day into the DWGM but when the gas stored reduces to below around 6,000 TJ (25%) this injection rate is expected to reduce substantially.

This is likely to result in gas shortfalls on moderate/high demand days during winter. These gas shortfalls could result in the use of voluntary or mandatory restrictions in gas use through the activation of the Gas Curtailment Guidelines and could have flow on impacts to the electricity supply system through curtailment of gas power generation.

9. Response (including Relief)

Response is the action taken immediately before, during and in the first period after an emergency to reduce the effects and consequences of the emergency on people, their livelihoods, wellbeing and property, on the environment and to meet basic human needs.

9.1. Control arrangements

The Secretary of DEECA is the Control Agency Officer in Charge[9] for a major energy emergency (refer page 52 of SEMP for CAOiC role statement). The Secretary (or State Controller – Energy (SC-E) if appointed) directs the response to a major energy emergency, including coordinating with other government agencies.

The Secretary (or SC-E, if appointed) may transfer control to any other officer from another agency, with consent of the officer in charge of that agency.

State Controller – Energy

The State Controller – Energy (SC-E) reports directly to the Secretary and is responsible for establishing and monitoring a management structure, as appropriate, to ensure effective control arrangements.

The role of a SC-E is defined in the SEMP Class 2 Controller role statement (refer page 75 – 76 of SEMP). In summary, the SC-E:

- develops and coordinates implementation of strategic plans in response to major emergencies
- works with the Emergency Management Commissioner to establish and chair the State Control Team or works alongside other state-tier controllers, if appointed
- participates in the State Emergency Management Team
- gives directions to Regional Controllers and/or Incident Controllers as necessary
- with the support of the State Control Centre Room Manager, oversees the operational functioning of the State Control Centre
- identifies current and emerging risks, or threats and implements proactive response strategies
- ensures the timely flow of relevant information between government and industry to inform planning by State Control Team, State Emergency Management Team and State Relief and Recovery Team
- facilitates transition to recovery at the state level, supported by the State Emergency Relief Coordinator and State Recovery Coordinator, and provide support at regional and local levels
- public information, community warnings and advice

The SC-E must keep the Emergency Management Commissioner informed about:

- the effectiveness of the control arrangements for the major energy emergency
- consequence management planning, implementation, and outcomes, in consultation with the State Consequence Manager
- the integration of recovery into the response arrangements, in consultation with the State Recovery Coordinator.

9.2. Application of State Emergency Management Priorities

Response decisions by the SC-E must consider and apply the State Emergency Management Priorities to guide all decisions during a response to any emergency. These priorities are defined within the SEMP (refer page 7 of SEMP). **Table 10** outlines how the state priorities are applied for major emergencies where there is a risk to or actual impact to electricity, gas or liquid fuel supplies.

[9] Throughout this sub-plan the use of the Secretary will be used in lieu of Control Agency Officer in Charge

TABLE 10: State Emergency Management Priorities – Energy

State Emergency Management Priorities	Application for Class 2 emergency (energy)
Protection and preservation of life and relief of suffering is paramount. Including the safety of emergency response personnel, and of community members including vulnerable community members and visitors/tourists	The protection of life always takes primacy during electricity or gas supply or liquid fuels disruption. Energy emergencies may pose a significant risk to safety, health and wellbeing of emergency response personnel and the community during and after an emergency. Safety information for emergency response personnel and the community about dangers, including the risk of electrocution from fallen power lines, hazardous smoke and gas leaks is a key priority. Coordination and dissemination of accurate, timely and fit-for-purpose impact data informs planning and delivery of health and wellbeing support, noting some people are particularly vulnerable to energy supply disruptions. Prolonged power outages are a significant risk to the health, safety, and wellbeing of life support customers whose life support equipment relies on electricity or gas to operate. Electricity distribution businesses have a legislative role in ensuring life support customers are informed of prolonged power outages and Department of Health legislative role to check on life support customers affected by prolonged power outages.
Issuing of community information and community warnings detailing incident information that is timely, relevant and tailored to assist community members make informed decisions about their safety.	DEECA will coordinate with AEMO, electricity/gas distribution businesses, ESV and owners/operators of infrastructure, transmission, and distribution assets to prepare and deliver information and community warnings. Community messaging will vary based on the location, type, and duration of the emergency.
Protection of critical infrastructure and community assets that support community resilience.	Disruptions of energy critical infrastructure and other assets that support individual livelihoods can compound the health and safety of communities, business, and industry. Agencies with a role in protecting critical infrastructure must prepare, test, and update their business continuity plans (as required under the EM Act 2013) to reduce the disruption caused by an energy emergency. Identification and management of energy disruption on critical infrastructure and community assets is a key focus of responding to an energy emergency. Timely and effective two-way information sharing supports critical infrastructure sectors and community owned/operated assets that support community resilience by: • identifying impact to inform prioritisation of restoration efforts by transmission and distribution businesses • providing accurate and reliable information about energy supply disruption, there duration and location enabling planning of business continuity arrangements and targeted relief activities. AEMO are critical in ensuring protection of electricity and gas supplies and implementing measures to maintain reliable supplies to Victoria Owners/operators of critical energy infrastructure are required under the EM Act 2013 to develop and implement a risk management plan and undertake works to ensure the protection of assets, including vegetation clearance to prevent ignition from powerlines.
Protection of residential property as a place of primary residence.	All Victorians including the emergency management sector, households, business, government, and industry have a shared responsibility to prepare for a disruption to electricity, gas, or liquid fuel supplies to reduce the likelihood of impacts from these emergencies.
Protection of assets supporting individual livelihoods and economic production that supports individual and community financial sustainability.	Services, business, and industry that depend upon power, gas and liquid fuel must ensure they prepare, test and update their business continuity plans to reduce the disruption that may be caused by an energy emergency. Timely and effective two-way information sharing supports the ongoing operation of businesses and assets that support livelihoods:

OFFICIAL

State Emergency Management Priorities	Application for Class 2 emergency (energy)
	• identifying business impact informs prioritisation of restoration efforts by energy industry • providing accurate and reliable information about energy supply disruption duration and location enables planning around business continuity plan implementation, and targeted relief activities.
Protection of environmental and conservation assets that considers the cultural, biodiversity, and social values of the environment.	Energy emergencies may influence and impact the natural environment. Use of hazardous materials effect cultural, biodiversity and social values of the environment.

9.3. Roles and Responsibilities Energy

Under the SEMP, DEECA is the control agency for major emergencies involving electricity, gas or liquid fuel emergencies. Activation of DEECA energy response arrangements occurs when one or more of the triggers in **Tables** 12, 13, 14 are met or exceeded.

As the operator of the electricity and gas systems and markets in Victoria, AEMO has a range of powers to intervene in energy markets to respond to risks to the security of energy supply and may utilise load shedding and gas curtailment. During an energy emergency, AEMO provides expert advice to DEECA about electricity and gas supply disruption; and coordinates and directs the gas and/or electricity industries to respond.

As the safety regulator of electricity, gas and pipelines, ESV provides safety and technical advice, including technical specialists to lead investigations, and to exercise emergency powers to make safe an emergency.

Services, businesses and industry that depend upon power, gas and liquid fuel are responsible for enacting their business continuity plans. Under the SEMP, other government departments (Department of Government Services, Department of Health, Department of Families, Fairness and Housing, Department of Jobs, Skills, Industry and Regions, and Department of Transport and Planning) have key roles to support critical infrastructure owners and operators for potential or actual major energy emergencies and the upstream or downstream dependencies disrupted. These government departments provide information on sector impacts to DEECA who coordinates restoration to inform planning for and delivery of sector-specific support and relief.

As the owners and operators of energy infrastructure and supply chains, the energy industry has the primary responsibility to respond to energy supply disruption and deliver repairs, restoration and reconnection to Victoria. The energy industry also provides subject matter experts to advise and support DEECA during major emergencies through the Energy - Emergency Management Liaison Officer.

Table 11 outlines what impacts are likely to occur from other major emergencies and which DEECA would support the response to risk and consequences to energy infrastructure, systems and/or supplies.

TABLE 11: Control arrangements

Hazard	Energy Supply Disruption	Control Agency
Cyber attack	Potential to interrupt services and cause physical damage to assets and systems	Department of Government Services (DGS)
Earthquakes	Potential to damage gas and liquid fuels pipelines, or electricity infrastructure causing supply shortfall and/or localised disruptions of energy supply.	Victoria State Emergency Service (VicSES)
Fires	Structure fires, bushfires and other fires can damage assets and infrastructure and disrupt	Country Fire Authority (CFA) Fire Rescue Victoria (FRV)

Hazard	Energy Supply Disruption	Control Agency
	supply chains across gas, electricity and liquid fuels (including retail)	Forest Fire Management Victoria (FFMVic)
Flooding	Potential to disrupt mining activities that may result in loss of power generation; can cause asset damage and disrupt access, road freight and supply chains	Victoria State Emergency Service (VicSES)
Hazardous materials	Incidents impacting energy infrastructure can include gas leakage and other hazardous materials.	Country Fire Authority (CFA) Fire Rescue Victoria (FRV)
Heatwave	A sustained period of high temperatures (heatwave) or a single day of higher-than-average temperature can increase electricity demand and reduce efficient generation and can disrupt gas production and transmission.	Emergency Management Victoria (EMV)
Landslide	Potential to cause damage to gas, electricity or liquid fuels infrastructure with localised or widespread impact	Victoria State Emergency Service (VicSES)
Pandemic	Potential to impact critical workforce availability across supply chains	Department of Health (DH)
Severe storms	Potential to cause extensive asset damage to electricity distribution or transmission network infrastructure, resulting in widespread power outages.	Victoria State Emergency Service (VicSES)
Warlike act or act of terrorism, hijack, siege or riot	Potential to cause infrastructure damage, supply chain disruption and/or harm to employees, with localised or widespread impact	Victoria Police (VicPol)
Maritime	Potential to disrupt the transportation of energy supplies delivered through Victorian ports.	Department of Transport and Planning (DTP)

9.4. State Control Centre – Energy Unit

DEECA is integrated within the State Control Centre as the Energy Unit (SCC-E Unit) and performs a range of roles within the Australasian Inter-Service Incident Management Systems at State, Region and Incident levels, where required. The linkages within the Australasian Inter-Service Incident Management System functions and the SCC-E Unit are dependent on the type of major emergency and whether a major energy emergency is proclaimed and/or if the Secretary appoints a SC-E. The SCC-E Unit integrates with AIIMS functions through:

- developing and delivering operational interventions such as Ministerial directions where required to augment other response measures being implemented by AEMO and industry
- ensuring coordination across multiple government and industry operations
- developing specialist communication packages for the Public Information Unit to inform advice, warning messaging and media products
- providing tailored intelligence and subject matter expert support on electricity, gas or liquid fuels to the State Controller, State Response Controller, State Control Team, State Intelligence Unit, State Planning Unit, State Consequence Unit, State Relief and Recovery Unit and other agencies command and Emergency Management Liaison Officers.

9.5. Concurrent emergencies

An energy emergency can occur concurrently with, or consequentially from, another major emergency (class 1, class 2 or class 3).

When an energy major emergency is concurrent with another emergency, the Secretary (or SC-E, if appointed) consults with the Emergency Management Commissioner and any other relevant control agency to agree and document effective and appropriate control arrangements. This can include coordinated but separate control arrangements, with multiple state tier controllers responsible for each emergency.

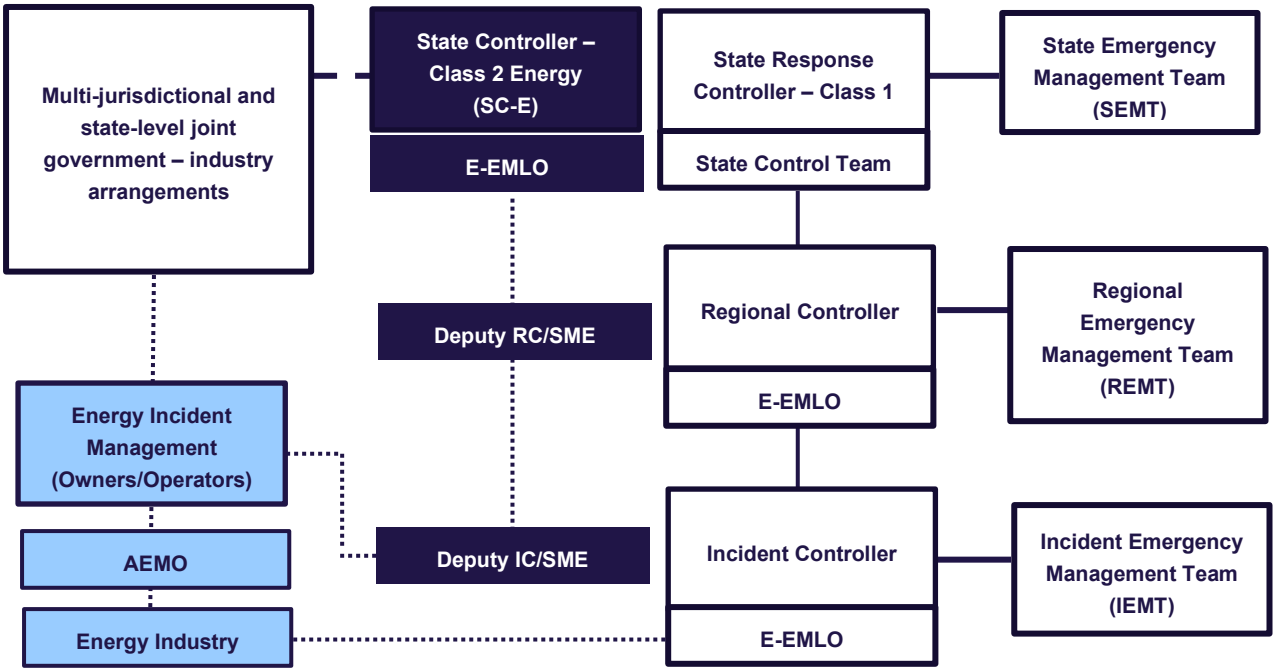
Where regional and incident tiers of control are established for another major emergency, energy subject matter experts can be embedded into those tiers, for example through:

- a deputy regional controller or deputy incident controller
- an Energy – Emergency Management Liaison Officer, industry representative or personnel from DEECA.

In the event of a localised or limited impact energy disruption concurrent with a major emergency, DEECA supports the relevant control agency through subject matter expertise, intelligence and reporting, and energy sector engagement and coordination (**Figure 4)**. Examples of a localised or limited impact energy disruption (or risk) concurrent with a major emergency may include:

- localised power outages affecting a relief centre established in relation to a major emergency
- closure or stock-out at local liquid fuel retailer due to reduced freight access as a result of fire, flood or storm, or workforce shortage due to pandemic
- risk to a gas pipeline or electricity transmission line from fire, flood or other natural hazards.

FIGURE 4: Example Control structure for concurrent emergencies



9.6. Operation of multi-jurisdictional and state-level joint government-industry arrangements

Energy emergencies are managed in accordance with established multi-jurisdictional and joint government-industry emergency management arrangements (**Figure 5)** as required.

Multi-jurisdictional and state-level joint government-industry arrangements are underpinned by agreements between Australian state and territory governments and industry about the management and response to energy disruptions.

OFFICIAL

These arrangements are delivered by governance bodies who oversee planning, preparedness and response for each sector (electricity, gas and liquid fuels).

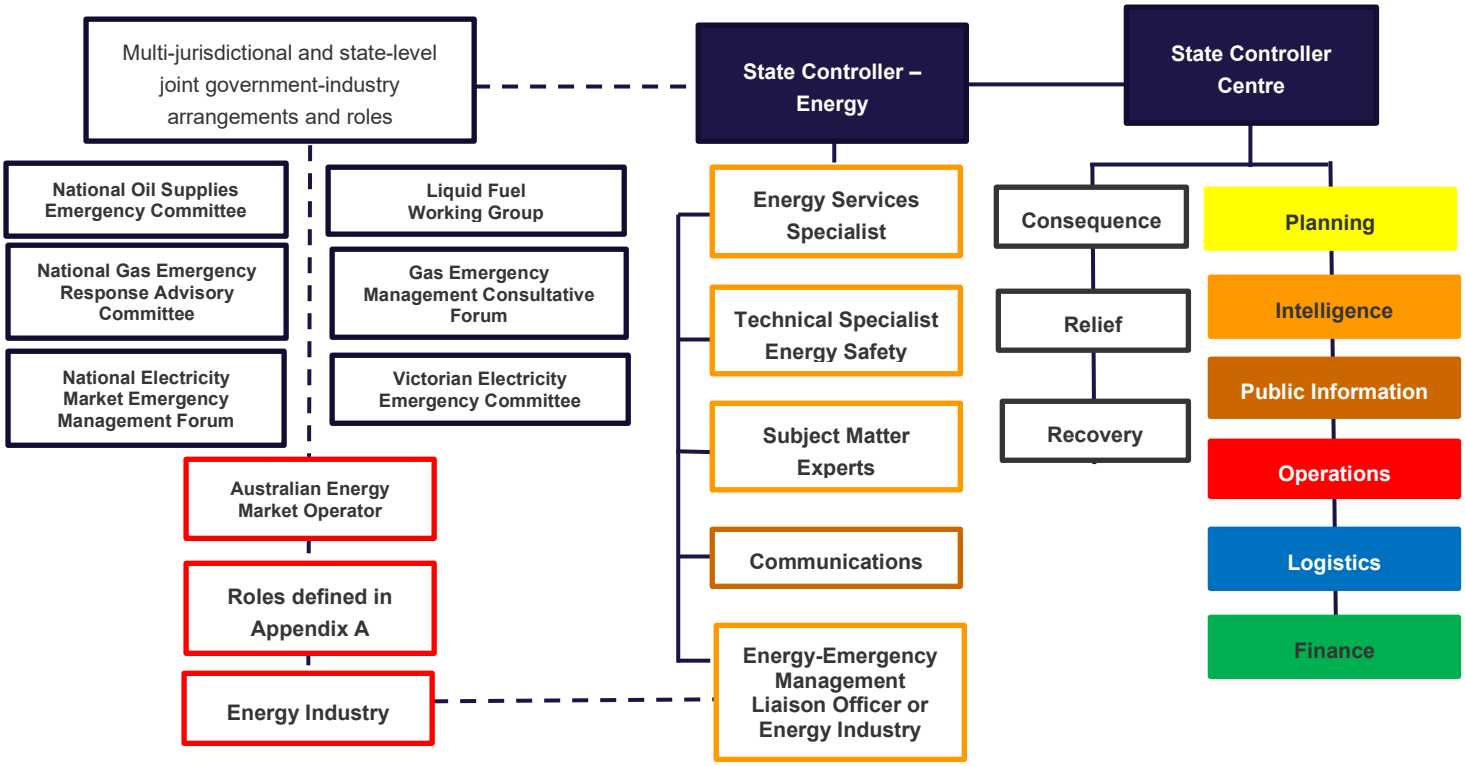
Multi-jurisdictional and state-level arrangements are independent of each other and can be activated concurrently. Mechanisms provide for information sharing and coordination between the two sets of arrangement for each energy sector.

These arrangements support timely and consistent intelligence gathering, risk assessment and information sharing, and enable more effective coordination of response, relief and recovery.

These arrangements complement the roles and responsibilities as defined in the EM Act 2013, the SEMP and this sub-plan.

See **Appendix A** for a summary of multi-jurisdictional and joint government-industry emergency management arrangements.

FIGURE 5: Example government-industry control structure



9.7. Market Interventions and Legislative Powers

There are a range of interventions and powers that can be exercised in response to an incident involving energy infrastructure or supply security. The powers can be enacted by the Minister for Energy and Resources (State), ESV, DEECA, AEMO and the Minister for Climate Change and Energy (Cth).

These powers generally allow for market interventions, preparedness arrangements and, voluntary or mandatory measures and directions to ensure the safe and reliable supply of energy. Consideration and implementation of these interventions occurs within the established multi-jurisdictional and joint government-industry emergency management arrangements.

9.8. Critical Infrastructure, community assets and essential services

All critical infrastructure, community assets and essential services rely on energy supply for ongoing delivery of their services to community. Interruption to energy supply can cause cascading effects, across multiple sectors simultaneously. These interdependencies between essential service sectors are a risk multiplier.

In the event of an energy disruption, the owners/operators of critical infrastructure, community assets and essential services implement their continuity and contingency plans. These sectors rely on timely and reliable information about the extent and expected duration of energy disruption to inform implementation of these plans.

During a major energy emergency:

- relevant support agencies (as defined in the SEMP Roles and Responsibilities) will liaise between the owners/operators of critical infrastructure, community assets and essential services and DEECA, as the Control Agency
- DEECA regularly seeks advice from relevant support agencies on the impact on critical infrastructure, community assets and essential services. DEECA shares this information with relevant energy sector entities, and provides advice to the energy sector about the application of the State Emergency Management Priorities to inform prioritisation of restoration of supply, within network, market, or system constraints, and
- DEECA works with the energy sector to share intelligence with relevant support agencies for response the about disruption risks, estimated time to restore, geographic impact, critical infrastructure impact, and restoration prioritisation.

9.9. Response to electricity major emergencies

Where an energy emergency is likely to result in demand exceeding supply (electricity), DEECA provides support to make public appeals for voluntary demand reduction based on advice from AEMO.

AEMO has a range of powers to intervene in the electricity market including market notices, use of the Reliability and Emergency Reserve Trader, and load shedding. AEMO may also suspend the energy market in a region under clause 3.14.3 of the National Electricity Rules.

To mitigate the risks of major electricity emergencies due to supply demand imbalance, AEMO forecasts demand and supply and signals to the market, through Lack of Reserve notices **(Table 12**), when a response is required to increase generation capacity or demand response is required.

TABLE 12: Lack of Reserve Levels

Lack of Reserve Level (LOR)	Definition	Example
LOR1	Victoria has enough supply to meet forecast demand, but reserves are forecast to fall below the LOR1 Trigger. As such, AEMO seeks additional reserves.	There is a risk that supply will not meet forecast demand if either: (i) the two largest generating units are lost (e.g., two of Loy Yang A's four units at 560 MW each); or (ii) both the largest generating unit and Basslink are lost (e.g., a Loy Yang A unit at 560 MW plus Basslink at 594 MW).
LOR2	Victoria has enough supply to meet forecast demand, but reserves are forecast to fall below the LOR 2 Trigger. As such AEMO seeks additional reserves.	There is a risk that supply will not meet forecast demand if either: (i) the largest generating unit is lost (e.g., a Loy Yang A unit at 560 MW); or (ii) Basslink is lost at 594 MW.
LOR3	Victoria is forecast to not have enough supply to meet demand. As such AEMO seeks additional reserves. An update will be provided, if additional reserves are not available, load shedding is expected.	Supply does not meet demand. Load shedding occurs (either manual or selective depending on available time).

An appeal for voluntary load reduction can be made by the Minister for Energy and Resources, or a government spokesperson on their behalf. Reducing electricity consumption during peak demand periods can potentially help to mitigate the need for 'load shedding'[10].

Load shedding is an action that AEMO can use as a last resort to maintain system security across the network and therefore prevent power blackouts or damage to infrastructure. This is a measure only

[10] Load shedding is where network businesses remove electricity demand by switching off areas of demand

used when all remaining reserves are exhausted, and it would typically occur during extreme weather events or when generation capacity is lost due to unforeseen circumstances.

AEMO works with the electricity network companies to rotate the outages across the network based on load shedding schedules developed by the Victorian Jurisdictional System Security Coordinator to minimise impact. Where possible, outages are rotated to ensure the same area is not impacted more than once.

The *Electricity Industry Act 2010* and the *Electricity Safety Act 1998* provide general powers of enforcement to ESV in relation to safety and other related obligations. Under the *Electricity Safety Act 1998* the Chairperson of ESV may do or give any direction they consider necessary to make an electricity emergency safe.

Under the *Electricity Industry Act 2010*, the Governor in Council may proclaim an electricity supply emergency if:

a) an event has occurred, or is about to occur, which may endanger an undertaking of a distribution company, a transmission company or a generation company or a person who supplies electricity to another person or materially affects the safe, economical or effective supply of electricity; or

b) the available supply of electricity is, or is likely to become, less than is sufficient for the reasonable requirements of the community.

The proclamation must be published in the Government Gazette and takes effect on the date of its publication; the Governor in Council may at any time revoke a proclamation.

Once this proclamation is made, the Minister for Energy and Resources may issue directions in accordance with the *Electricity Industry Act 2010* necessary to:

a) protect any undertaking of a distribution company, a transmission company or a generation company or a person who supplies electricity to another person; or

b) ensure the safe, economical or effective supply of electricity; or

c) ensure that the available electricity is fairly distributed to the community; or

d) increase the available supply of electricity; or

e) regulate the use of the available supply of electricity, having regard to the needs of the community

Electricity Thresholds and Triggers for Major Emergencies

The management of a potential or actual major electricity emergency is guided by forecasts, thresholds and triggers that determine the level of response and actions required to manage these conditions (**Table 13)**. DEECA will liaise with the AEMO, energy industry owners/operators, State Control Centre and other agencies to determine when a threshold has been met and will determine when arrangements escalate from readiness to response.

TABLE 13: Electricity Thresholds and Triggers Major Emergency

MAJOR EMERGENCY					
FORECAST			EVENT		
Weather	Reserves, Supply and Demand	Other Conditions	Power Outages	Lack of Reserve	Other Conditions
"Severe" heatwave warning issued by the BOM that affects major population centres[11] outside metropolitan Melbourne: • three or more major population centres; and/or • metropolitan Melbourne. "Extreme" heatwave warning issued by the BOM that affects the major population centres outside the metropolitan Melbourne: • two or more major population centres; and/or • metropolitan Melbourne. A 'Health Alert: Heat Health Warning' is issued by the Chief Health Officer that affects major population centres outside metropolitan Melbourne: • three or more major population centres; and/or • metropolitan Melbourne. Forecast of Fire Behaviour Index of 75 plus in the Australian Fire Danger Rating System (high end of Extreme or Catastrophic) Forecast wind gusts over 110km/h in non-alpine, and 130km/h in Alpine areas Two days with maximum forecast temperatures more than 45° c in any weather district Two days with maximum forecast temperatures more than 38° c in the central weather district	AEMO issues a forecast Lack of Reserve 3 market notice AEMO notifies DEECA of potential for load shedding following assessment of reserve conditions An event is about to occur which may endanger an undertaking of a distribution business, a transmission business or a generation business or a person who supplies electricity to another person or materially affects the safe, economical or effective supply of electricity	A gas supply shortage threatens the operation of gas-fired power stations High profile event, threat or natural disaster Any incident with impact for Victoria that is classified as a Level 4 or Level 5 incident under the PSEMP Any incident considered likely to escalate to a major energy emergency	Generation, transmission or distribution equipment failure or disruption (including supply chain disruption) causing outages for: • actual unplanned off supply more than 20,000 customers, in a single distribution business' distribution zone for greater than 30 minutes in a single distribution zone • unplanned transmission or distribution equipment failure resulting in a single credible contingency event which reduces redundancy for more than 20,000 customers for more than 12 hours • more than 10,000 customers with a return to service estimated to be more than 72 hours • critical infrastructure • industry resulting in, or likely to result in, significant commercial or industrial impact	AEMO issues a market notice for actual Lack of Reserve 3 conditions The available supply of electricity is, or is likely to become, less than is sufficient for the reasonable requirements of the community	Any incident attracting or likely to attract significant media or public interest with a risk of reputational damage to the State A large and complex incident that has the potential to have or is having significant adverse consequences to the safety and wellbeing of the Victorian community or a part of the Victorian community An event has occurred which may endanger an undertaking of a distribution business, a transmission business or a generation business or a person who supplies electricity to another person or materially affect the safe, economical or effective supply of electricity Proclamation of an electricity supply emergency under the *Electricity Industry Act 2000*

[11] Bendigo, Mildura, Shepparton, Albury-Wodonga, Warrnambool, Traralgon

OFFICIAL

9.10. Mutual Aid Arrangements

There is an agreed process between Victorian electricity distribution businesses to obtain, where possible, short-term assistance in the form of personnel, equipment, materials, and other related services outside the area that a distribution business operates.

The primary objective of this arrangement is to facilitate rapid, short-term deployment of assistance following an event. Mutual aid may be requested when an incident occurs, and the distribution business has exhausted or is nearing exhaustion of all its available resources.

Industry can request government support where a distribution business has exhausted or is nearing exhaustion of all its available resources including mutual aid. During widespread natural disasters, this may include specialist support for example logistics support from emergency agencies to enable access to energy infrastructure and assets.

9.11. Response to gas major emergencies

AEMO has a range of powers to intervene in the gas market. This includes emergency provisions under the National Gas Rules, and the use of market notices.

AEMO must ensure that flows of gas from the transmission network into distribution networks and interconnecting pipelines are adequate to meet demand and maintain minimum pressures in the Declared Transmission System. Where there is a threat to or breach of system security, AEMO may give directions to 'curtail'[12] gas load.

AEMO provides planning and forecasting information to support efficient decision-making and long-term investment in Australia's gas markets and infrastructure services. AEMO publish the Victorian Gas Planning Report which informs market participants and government about gas transmission capacity changes that impact security and reliability in the Declared Transmission System.

AEMO runs year-round, 24-hour operations of the Declared Transmission System to ensure its safe and effective operation and schedules the Declared Wholesale Gas Market.

Gas pipeline operators, storage, distributors, and producers operate and constantly monitor equipment and systems to ensure safe and effective operations.

When a supply shortfall is identified by AEMO, they will issue a range of measures to limit gas usage may include:

- market notices and ad hoc market schedules
- directions for injection or restriction on withdrawals
- public appeal by the Minister for Energy and Resources for a voluntary reduction in gas consumption
- Governor in Council may invoke the emergency provisions of the *Gas Industry Act 2001*allowing the Minister for Energy and Resources to issue directions to restrict consumption
- AEMO will recommend to DEECA that the Minister for Energy and Resources issue directions for mandatory load curtailment to market participants, market customers and gas retailers, who will then be directed to instruct their customers to limit gas usage.

Curtailment directions are only issued as a last resort after due consideration of alternative sources of gas supply and are limited to the extent required to maintain or restore system security.

The *Gas Industry Act 2001, Gas Safety Act 1997* and *Pipelines Act 2005* provide powers to ESV and the Minister for Energy and Resources (or delegate) in relation to emergencies, safety and other regulatory obligations.

[12] Consumers are asked or directed to reduce their gas consumption or demand through voluntary or mandatory measures as per the Gas Curtailment List.

Under the *Gas Industry Act 2001*, the Governor in Council may, on the advice of the Minister for Energy and Resources, proclaim a gas supply emergency where supply of gas is or is likely to be insufficient for the reasonable requirements of the community. Under the *Gas Industry Act 2001*, and while a proclamation is in force, the Minister may give any directions necessary to:

(a) ensure the safe and secure supply of gas; or

(b) to regulate the use of the available supply of gas.

Decisions about which intervention, direction, or powers are best utilised for the management of a gas emergency will be supported by the Gas Emergency Management Group (**Appendix A**).

Gas Thresholds and Triggers for Major Emergencies

Arrangements for a major gas emergency are determined through a range of forecast and actual thresholds and triggers (**Table 14)** that inform levels of response and actions required to manage these conditions.

Advice from AEMO and gas owners/operators on when actual thresholds are likely to or have been exceeded is provided to DEECA who activates readiness or response arrangements.

Activation of response arrangements may include appointment of a SC-E by the Secretary.

TABLE 14: Gas Major Emergency Thresholds and Triggers

MAJOR EMERGENCY					
FORECAST			EVENT		
Weather	Reserves, Supply and Demand	Other Conditions	Gas Outages	Gas Supply Disruption	Other Conditions
Two days with maximum forecast temperatures of less than 12° c for the Melbourne metropolitan area One day with a maximum forecast temperature of less than 10° c for the Melbourne metropolitan area	AEMO issues a forecast a market notice warning of insufficient gas supply to meet demand (e.g threat to system security notice) AEMO notifies DEECA of potential for gas curtailment following AEMO's assessment of the forecast gas supply and demand conditions An event or outage is about to occur, such as urgent outage at a major gas supply facility or on the DTS or a pipeline strike that may endanger facility employees and/or the public resulting in a shutdown or capacity reduction that will result in a supply shortfall	An electricity supply outage threatens the operation of gas supply infrastructure, that may require the prioritisation of diesel for onsite power generation to prevent a major reduction in gas supply High profile event, threat or natural disaster Any incident that results in an activation of the NGERAC Interruption to Supply process Any incident considered likely to escalate to a major energy emergency	More than 1,000 customers are impacted by loss of gas supply Substantial risk of serious injury or death Significant commercial or industrial impact Immediate threat to significant critical infrastructure or facilities including hospitals, major industry and/or other utilities	Loss of supply form a major injection point or unplanned isolation of a significant transmission pipeline jeopardising the integrity of the principal network. Any incident leading to gas supply curtailment. Proclamation of a gas supply emergency under the *Gas Industry Act 2001*	Any incident attracting or likely to attract significant media or public interest, with a risk of reputational damage to the State. A large and complex incident that has the potential to have or is having significant adverse consequences for the safety and wellbeing of the Victorian community or a part of the Victorian community. Degree of complexity requires more substantial organisational structure.

OFFICIAL

9.12. Life Support Customers (Electricity and Gas)

Response to major emergency is underpinned by the State Emergency Management Priorities (**Table 11**) which guide decisions during a response to any emergency. These priorities include the protection and preservation of life and relief of suffering, including the safety of vulnerable persons who are dependent on electricity or gas for medical reasons.

Electricity

Section 2.1 of the *Electricity Distribution Code of Practice* defines the term 'life support customer', also referred to as 'power dependent customer' as those customers who are registered as dependent on electricity for medical reasons. Distribution businesses are responsible for providing all customers with an emergency contact number and regular updates on the status of a power outage on their website.

The 'Power Outage Notification Protocol' (the Protocol) is an agreement between the Department of Health and Victorian electricity distribution businesses. The Protocol sets out the forecast conditions in which preparedness for joint activation by Department of Health and Department of Families, Fairness and Housing will occur, triggers for activation and the manner and format of information required to be supplied to Department of Health and Department of Families, Fairness and Housing.

Activation of the Protocol is triggered when a widespread supply event occurs when distribution businesses form the view that **both**:

- More than **20,000 customers** from a single distributor are without power,
- **and** those customers are expected to remain without power for more than **24 hours**.

An initial list of life support customer is sent by the distribution businesses to Department of Health and Department of Families, Fairness and Housing as soon as the Protocol is activated after which they are required to provide a list of life support customers every 12 hours until power is restored to all customers. The Protocol provides the format and details of what information is to be provided to Department of Health and Department of Families, Fairness and Housing to ensure prioritisation of customers for welfare checks. It is the responsibility of life support customers to develop an emergency plan, in collaboration with their general practitioner and/or carer on the arrangements required in response to a power outage.

Activation may occur outside of these triggers at the discretion of either Department of Health and Department of Families, Fairness and Housing where there are potential or actual conditions that are likely to result in significant power disruption. When triggers for a major emergency are met, information is requested to lessen or prevent a serious threat to the life, health, or safety of any individual, or to public health or safety.

Deactivation of the Protocol occurs once distribution businesses advise Department of Health and Department of Families, Fairness and Housing that all life support customers impacted by the incident have had electricity restored.

Gas

Section 4A of the Gas Distribution System Code of Practice defines a Life Support Customer where the customer requires supply of gas at the premises for the purposes of operating the life support equipment for medical reasons.

Customers that have a medical requirement to consume gas that are registered on the gas Life Support Register, through their distribution business, are prioritised above all other residential gas customers but below essential health services for mandatory gas curtailment.

These customers will be managed in accordance with the State Emergency Management Priorities (**Table 11**) and the implementation of AEMO's emergency gas procedures.

9.13. Response to liquid fuel major emergencies

A range of government and industry response mechanisms to manage liquid fuel disruptions are available, depending on the duration and extent of the disruption. In the absence of a major

emergency, the policy of the Victorian Government is to allow the liquid fuel industry, customers and consumers to manage the fuel supply and demand balance without government intervention.

Industry uses a variety of production, supply chain management and pricing mechanisms to manage supply and demand and respond to potential and actual fuel supply issues.

More direct response measures include voluntary demand reduction communications or the implementation of a retail rationing scheme using either Commonwealth or Victorian legislation. Use of emergency legislative powers is considered a process of last resort.

The *Fuel Emergency Act 1977 (Vic)* provides for a proclamation that a state of emergency (**Table 15**) exists when the Governor in Council is satisfied that an event has occurred or is imminent "whereby any kind of fuel is or is likely to become unavailable to meet the reasonable requirements of the community". During a period of emergency, the Minister for Energy and Resources can "regulate and direct any service" relating to the production, supply, distribution, sale, use or consumption of fuel.

Under the *Liquid Fuel Emergency Act 1984 (Cth),* the Governor General may declare a national liquid fuel emergency in the event of an actual or likely national fuel shortage. Victoria must be consulted alongside all the other states and territories before a declaration is made. The *Liquid Fuel Emergency Act 1984* gives the Minister for the Industry, Energy and Emissions Reduction (Cth) powers in an emergency to control industry held stocks of crude oil and liquid fuels produced by Australian refineries, and fuel sales across Australia.

During a national liquid fuel emergency, Victoria aligns its approach, including for prioritising essential users, with the national approach. Essential users are defined in the *Liquid Fuel Emergency Act 1984* and the Commonwealth's Essential Users Determination 2019.

Prioritisation of essential users does not mean that there will be no fuel available for other users. Essential users typically represent only a relatively small portion of overall fuel demand. It is anticipated that under most scenarios, other users will still be able to access some fuel, albeit at reduced quantities. Business continuity planning by entities not included under the determination should take this into account.

TABLE 15: Liquid Fuels Emergency Thresholds and Triggers

MAJOR EMERGENCY			
FORECAST		**ACTUAL**	
Liquid Fuel Supply Disruption	**Other Conditions**	**Liquid Fuel Supply Disruption**	**Other Conditions**
Imminent threat to supply for 'essential fuel users' and the broader community Increased risk or perceived risk of significant supply disruption Short-duration supply disruption with small-moderate impact on fuel users, with potential to escalate level of impact or duration. Incident or situation with likely risk to domestic supply chain, including retail network, with significant adverse impact on the Victorian community, or part of the Victorian community.	NOSEC or LFWG convened in response to conditions or incident. A prolonged electricity or gas supply outage likely to affect the operation of a major refinery or fuel terminal. A global market, supply chain or geo-political event likely to materially impact either refinery operations or the volumes of crude oil or refined fuel that can be imported into Victoria. Any 'limited' level liquid fuel event with potential to escalate into a major emergency.	Material, widespread and/or prolonged reductions in available supply for 'essential fuel users'[13] and the broader community. A moderate supply disruption with prolonged duration Issuing of a NOC Red Traffic light that is likely to be escalated, or a Black Traffic Light for Melbourne Airport. Substantial imbalances in the markets affecting primary grade fuel(s), such as diesel, petroleum and/or jet fuel A catastrophic plant failure of a vital liquid fuel critical infrastructure asset in Victoria Incident or situation has or is likely have impact to global or domestic supply chain and/or critical infrastructure assets, including retail network, with significant impact on the Victorian community, or part of the Victorian community.	Multi-jurisdictional impacts, including activation of the National Liquid Fuel Emergency Response Plan. Incident response requires utilising provisions of the *Fuel Emergency Act 1977.* A large and complex liquid fuel incident with potential or actual significant adverse consequences for the safety and wellbeing of the Victorian community or a part of the Victorian community. Degree of complexity requires more substantial response structure. Any incident attracting or likely to attract significant media or public interest, with a risk of reputational damage to the State.

[13] 'Essential fuel users' are defined in the Liquid Fuel Emergency (Activities—Essential Users) Determination 2019.

OFFICIAL

9.14. Consequence Management

Consequence management is the coordination of agencies responsible for managing or regulating services or infrastructure which are or may be affected by a major emergency. Consequence management supports strategic decision-making before, during and after a major emergency. The importance of supporting long-term decision-making following a major emergency, including recovery activities, should be a priority.

The Emergency Management Commissioner is responsible for the management of consequences from major emergencies. The Emergency Management Commissioner fulfils this role through the appointment of the State Consequence Manager. This role is responsible for providing strategic and management advice about actual, emerging and cascading consequences before, during and after a major emergency and working with the State Emergency Management Team and Regional Emergency Management Teams, if activated to ensure a whole-of-government approach to the management of these consequences.

Any major emergency has a risk upon energy infrastructure, assets or systems, these consequences my result in impacts to the safe and reliable supply of energy to the state, however, will be dependent on duration and the extent of consequences. Each critical infrastructure sector will have its own consequences as a downstream dependency on electricity, gas or liquid fuels.

The Secretary (or a SC-E is appointed), the SCC-E Unit will support the State Consequence Manager in managing consequences of potential or actual impacts to energy.

To achieve this, DEECA works with industry and support agencies to:

- coordinate identification of consequence, including on critical infrastructure sectors
- support the State Consequence Manager on state consequence management planning
- coordinate communications to affected communities and stakeholders
- work with other critical infrastructure sectors on the priority site restoration based on assessment of consequences to the provision of essential services.

9.15. Public Information and Warnings

Public information and warnings ensure that the community has the information it needs to make informed choices about their safety and to take responsibility for their own recovery.

Under the SEMP, the Emergency Management Commissioner or the State Response Controller must ensure that warnings are issued. Victoria has an integrated warning system which provides information and warnings to community through a range of channels.

The Emergency Management Commissioner works with SCC-Energy Unit to develop appropriate communication and engagement plans in relation to major energy emergencies.

The Secretary (or SC-E if appointed), may seek support from the Emergency Management Joint Public Information Committee to ensure state level messages from all agencies involved in the management of emergency response, relief and recovery are included in the key messages to the public.

Energy Emergency Advice and Messaging

DEECA uses a suite of pre-prepared communication materials in response to forecasts and events with a risk of energy supply disruption, for example bushfires, heatwave and storms. The aim of this communication is to support industry to reduce electricity and gas consumption prior to supply shortfalls and advise the community about temporary changes to supply during incidents.

DEECA also delivers a broad public information campaign to educate Victorians prior to summer about how severe weather events can lead to unplanned outages and the importance of reducing electricity consumption during instances when power infrastructure may be under extreme pressure, and to mitigate the risk of load shedding events.

The goal of these education campaigns is that Victorians proactively reduce energy consumption on hot days or in heatwave events and that vulnerable Victorians proactively seek support or are supported during load shedding events and power outages.

ESV in addition to providing advice on reducing energy consumption also disseminate energy safety messaging in response to major emergencies.

Energy Industry Communications

Under the Electricity Distribution Code of Practice, distribution businesses are required to issue information to customers about power outages and expected restoration times across their area, within a set time frame, particularly power dependent customers.

The Single Industry Spokesperson Protocol for Electricity (2019) is designed to ensure media, customers and the public receive coordinated and timely communication on agreed key messages that give clear advice about the status of a widespread electricity emergency.

The Single Industry Spokesperson Protocol for Gas (2019) is an agreement made between government and industry. Under this agreement, AEMO will speak on behalf of the Victorian gas supply industry during widespread and prolonged gas shortages. The Single Industry Spokesperson Protocol only activates for major gas emergencies of Level 4 and above.

The Australian Institute of Petroleum Communications Protocol (2021) on major fuel supply situations or disruptions is a framework for notifications, reporting and coordination of emergency response between and industry and the Victorian Government. The protocol is activated via the Victorian Liquid Fuels Working Group, as soon as practicable, if a known fuel supply situation is or will likely have visible implications for fuel users in Victoria.

9.16. Relief

Relief is the provision of assistance to meet the essential needs of individuals, families and communities during and in the immediate aftermath of an emergency.

The delivery of relief will be considered and guided by the principles of relief coordination as outlined in the SEMP:

- emergency-affected communities receive essential support to meet their basic and immediate needs, including support payments for households, allocation of back up generation and relief centres
- relief assistance is delivered in a timely manner, in response to emergencies and where there is a forecast of prolonged energy disruption
- agencies communicate clear, relevant, timely and tailored information and advice to communities about relief services through multiple, appropriate channels
- relief promotes community safety and minimises further physical or psychological harm
- relief services recognise community diversity
- relief is adaptive, based on continuing assessment of needs, including ongoing assessments where it is likely that prolonged energy disruptions
- relief supports community responsibility and resilience, including advice on how to prepare for prolonged power or gas outages
- relief is well-coordinated, with clearly defined roles and responsibilities
- relief services are integrated into emergency management coordination efforts
- relief is coordinated and delivered as close to the community as possible, supported by regional and state-based activities as required.

Specific roles and responsibilities for delivery of relief services and coordination are set out in SEMP Relief Roles and Responsibilities. Tiers of relief co-ordination are assigned as follows:

OFFICIAL

- Emergency Recovery Victoria performs the role of state relief coordinator and is supported by a range of organisations
- Department of Families, Fairness and Housing are responsible for regional relief coordination
- Municipal Councils coordinate relief at local government level and establish relief centres or community hubs at the request of the Incident Controller

The Emergency Management Commissioner may appoint a State Emergency Relief Coordinator who, in consultation with the State Relief and Recovery Team, is responsible for coordinating state-tier provision of essential and urgent assistance to individuals, families and communities affected by a major energy emergency.

While responsibilities for activation of relief rest with the Incident Controller, Regional Controller or State Response Controller, as the control agency for major energy emergencies the SCC – Energy Unit performs an overarching role for relief where the Secretary (or SC-E, if appointed) may determine it is necessary to deliver relief activities for those residential and business customers affected by prolonged power outages. This may include the SCC-Energy Unit liaising with energy distribution businesses to monitor industry-led relief activities, and coordinate priority restoration that best meets the essential needs of the community.

Early relief for households affected by prolonged electricity or gas disruptions are coordinated between DECCA, State Emergency Relief Coordinator, State Relief and Recovery Team, energy distribution businesses and where appropriate Municipal Relief and Recovery Coordinator's.

Transition to Recovery

As response to an emergency decrease, agencies make decisions regarding recovery priorities by working cooperatively to share information, jointly plan and execute recovery activities.

The SEMP outlines the process and roles and responsibilities for transition from relief to recovery, including short-term recovery activities that should be considered in the transition plan.

The Secretary (or the SC-E, if appointed) will work with the State Recovery Coordinator to develop transition plans for recovery. Transition will be to the relevant Recovery Lead Agency, with support from Recovery Support Agencies.

Key considerations in determining when and how response will transition to recovery include:

- the extent to which concurrent emergencies (e.g., storm events) are impacting communities
- the extent of impact on communities, which may determine whether a prolonged transition period needs to be implemented
- the extent and known level of loss and damage associated with the emergency
- the extent of emergency relief required by affected communities.

During a prolonged energy supply disruption, the Secretary (or SC-E if appointed) will ensure that the transition to recovery considers the requirements of affected communities, including any temporary support that has been put in place to assist with essential needs.

During prolonged energy disruptions, transition to recovery will be phased, with the transition to recovery planning accounting for ongoing response in some regions or areas, and recovery in others. Controllers and Recovery Coordinators at relevant incident, regional and State tiers will agree on the timing and phasing of transition and who will be responsible for recovery activities.

10. Recovery

Recovery means the assistance of persons and communities affected by emergencies to achieve a proper and effective level of functioning. Emergency relief and recovery activities integrate with emergency response activities and commence as soon as the impact and consequences of the emergency are anticipated.

Specific roles and responsibilities for delivery of recovery coordination and recovery activities are set out in SEMP Recovery Roles and Responsibilities. These arrangements align with the following recovery principles that can be adapted to meet the needs of people and communities affected:

- Understanding the context within specific affected communities and how recovery activities can be tailored to a community's history, values and dynamics
- Recognising that recovery is complex, and dynamic based on the nature of the emergency and the context of the community affected
- Use a community-led approach placing community at the centre and allowing them to be responsive and flexible, are engaged and supported to move forward
- Coordinate activities through planning, coordinate and using an adaptive approach between community and partner agencies based on evolving assessment of impacts and needs
- Recognise and build capacity and resilience for individuals, communities, and organisations

Under the SEMP, DEECA is the recovery lead agency for built environment (energy services). DEECA liaises with industry planning for reinstatement and return to reliable supply, while considering the circumstances of each community and how energy can become more resilient. A range of mechanisms can be used to tailor recovery to the context of community resilience including:

- Micro-grids and community batteries that provide electricity to isolated communities when the main distribution network has been impacted
- Solar panels and batteries allow properties to have an additional power supply during outages to the main electricity network
- Underground powerlines to prevent fire ignition and mitigate the impact from storms

Box 2 provides a case study of the small town of Mallacoota in far East Gippsland who were devastated and isolated when the 2019/2020 bushfires impacted the town and how the community spirit drove recovery towards building resilience through an alternative power supply.

Box 2: Mallacoota Community Battery Case Study

In 2019/2020 the Black Summer bushfires impacted the small coastal town of Mallacoota in far East Gippsland. Not only were there significant impacts to homes, businesses and community facilities the town was left isolated when they were unable to access the one road in and out of the town. In addition, with many powerlines surrounding Mallacoota damaged the township were left without electricity, telecommunication, fuel and other essential services.

Electricity to Mallacoota is connected to the electricity network by a single powerline running hundreds of kilometres through dense bushland and national park. Historically, Mallacoota has been impacted by power outages caused by tree or animals disrupting supply.

As part of the town's recovery, the town worked with recovery agencies and AusNet Services to build a large battery storage system that provides electricity in response to an electricity disruption and has the capacity to provide electricity for up to three days.

Mallacoota Community Battery



Credit AusNet Services

Appendix A: Multi-jurisdictional and State-level joint government-industry arrangements

Electricity

Electricity	Arrangement	Description
State-Level	Victorian Electricity Emergency Committee	**Victorian Electricity Emergency Committee** participants include electricity distribution, transmission and generation businesses, DEECA (Chair), ESV, Victorian emergency services, Victoria Police, AEMO, Emergency Management Victoria, Department of Families, Fairness and Housing, Department of Health and other government agencies. The Committee can be convened concurrently with National Electricity Market Emergency Management Forum to manage any Victorian specific elements of an electricity emergency. This committee is not convened in emergencies but meets twice a year.
	Victorian Electricity Emergency Communication Protocol	The **Victorian Electricity Emergency Communications Protocol** ensures timely sharing of information between Victorian Electricity Emergency Committee members when responding to an actual or likely energy emergency event. Triggers for activation of the protocol include actual or forecast events that threaten the power system.
	Single Industry Spokesperson Protocol for Electricity in Victoria (SISP)	The Single Industry Spokesperson Protocol participants include: AEMO, CitiPower and Powercor, AusNet Services, Jemena, United Energy, ESV and DEECA (usually represented by the Communications Specialist). The Protocol ensures coordinated and timely agreed key messages and give clear advice about the status of a widespread electricity emergency and contains relevant triggers, procedures, policies, roles, responsibilities, contact details and draft media materials
Multi-Jurisdictional	National Electricity Market Emergency Management Forum	The **National Electricity Market Emergency Management Forum** is chaired by AEMO and consists of representatives from industry and state and federal governments. It is charged with reviewing and improving the emergency plans and procedures that apply during power system emergencies, including the Power System Emergency Management Plan. For a dual energy multi-jurisdictional event impacting both electricity and gas supply, the National Electricity Market Emergency Management Forum will coordinate with the National Gas Emergency Response Advisory Committee.
	Memorandum of Understanding on the Use of Emergency Powers and National Electricity Market Emergency Protocol	The **Memorandum of Understanding (MOU)** and the National Electricity Market Emergency Protocol were developed by the Jurisdictions, that is, South Australia, Victoria, New South Wales, Queensland, Tasmania, and AEMO, to co-ordinate actions b**etween** individual state legislation to manage power system security emergencies.
	Power System Emergency Management Plan (PSEMP)	Developed by AEMO in consultation with Na**tional** El**ectricity** Ma**rket,** state and territory government and industry. **The Power System Emergency Management Plan** facilitates a coordinated approach to National electricity market power system incidents. The Plan is the single point of reference for the electricity industry, government and stakeholders and provides the emergency management arrangements for the power system.
Defined Roles	Jurisdictional Responsible Officer (JRO)	The Jurisdictional Responsible Officer advises the Victorian government of electricity emergency risks and events and participates in the Victorian Electricity Emergency Committee and the National Energy Market Emergency Management Forum.
	Jurisdictional System Security Coordinator (JSSC)	The Jurisdictional System Security Coordinator advises government on the use of electricity emergency powers and develops the manual load shedding table with advice from the Victorian Electricity Emergency Committee Technical Working Group and participates in the National Energy Market Emergency Management Forum.

OFFICIAL

Electricity	Arrangement	Description
	Jurisdictional Designated Officer (JDO)	The Jurisdictional Designated Officer represents Victoria on the National Electricity Market Emergency Management Forum.

TABLE 15: Electricity multi-jurisdictional and state-level joint government-industry arrangements

Gas

Gas	Arrangement	Description
State-Level	Gas Emergency Protocol	Developed by AEMO in consultation with the Victorian Government and the gas industry, the **Gas Emergency Protocol** consists of a set of procedures regarding response, responsibilities, and communication requirements in the event of a Victorian gas emergency. The Protocol also includes the Gas Load Curtailment, Gas Rationing and Recovery Guidelines that provide detail of the priorities, principles, considerations, measures and processes that AEMO will apply when curtailment is required, and for the rationing and recovery of gas supplies. The Protocol can be convened concurrently with the National Gas Emergency Response Advisory Committee to manage any Victorian specific elements of a gas emergency.
	Gas Emergency Management Consultative Forum	The **Gas Emergency Management Consultative Forum** is a planning and coordination forum of industry representatives to consider issues relating to the effective management of emergencies related to the Victorian gas system. Jointly convened by ESVV and AEMO. The Committee meets twice annually and holds an exercise every year.
	Victorian Gas Emergency Communications Protocol	The **Victorian Gas Emergency Communications Protocol (2020)** has been developed by AEMO, government agencies, industry, and emergency services. The Protocol outlines the process for information sharing between Victorian Gas Emergency Consultative Forum members in preparation for and response to gas emergencies impacting the state of Victoria.
	Single Industry Spokesperson Protocol for Gas in Victoria **Gas Emergency Management Group**	**The Single Industry Spokesperson Protocol** participants ensures coordinated and timely agreed key messages and give clear advice about the status of a widespread gas emergency. The Protocol contains relevant trigger, procedures, policies, roles, responsibilities, contact details and draft media materials.
Multi-Jurisdictional	National Gas Emergency Response Advisory Committee	The **National Gas Emergency Response Advisory Committee** is a planning and coordinating forum made up of industry, federal and state government representatives. The Committee facilitates efficient and effective communication across industry and government during major natural gas supply shortages with the intent of minimising significant consequences for the community. The Committee meets twice a year. For a dual energy multi-jurisdictional event impacting both electricity and gas supply, the National Energy Market Emergency Management Forum may coordinate with the National Gas Emergency Response Advisory Committee.
Defined Roles	Jurisdictional C**ontact** Officer	The Jurisdictional Contact Officer represents Victoria on the National Gas Emergency Response Advisory Committee.

TABLE 16: Gas multi-jurisdictional and state-level joint government-industry arrangements

Liquid Fuels

Liquid Fuels	Arrangement	Description
State-Level	Liquid Fuels Working Group	The **Victorian Liquid Fuels Working Group** comprises DEECA, the Australian Institute of Petroleum, Viva Energy, ExxonMobil, Ampol and BP. Per the Australian Institute of Petroleum Communications Protocol (2021), DEECA may convene the Liquid Fuel Working Group where there is a significant disruption to supply for fuel users. This forum provides an ongoing two-way exchange of information and advice between government and industry on the expected extent of the

Liquid Fuels	Arrangement	Description
		disruption, impacts and consequences, and what the most effective response mechanisms would be. The Liquid Fuel Working Group can be convened concurrently with the National Oil Supply Emergency Committee to manage any Victorian specific elements of a national liquid fuel emergency.
	Australian Institute of Petroleum Communications Protocol (2021)	The **Australian Institute of Petroleum Communications Protocol (2021)** on major fuel supply situations or disruptions is a framework for notifications, reporting and coordination of emergency response between industry and the Victorian Government. The Protocol is activated via the Liquid Fuel Working Group, as soon as practicable, if a known fuel supply situation is or will likely have visible implications for fuel users in a jurisdiction(s).
Multi-Jurisdictional	National Oil Supplies Emergency Committee	The **National Oil Supply Emergency Committee** consists of Commonwealth, State and Territory government representatives who in cooperation with the liquid fuel industry respond to widespread national fuel shortages or emergencies. The Committee provides relevant advice and information on issues confronting national supply of crude oil and petroleum products and reports annually or as required to the Senior Committee of Officials on operations and the state of readiness.
	National Operating Committee for Jet Fuel Assurance	The **National Operating Committee for Jet Fuel Assurance** is a voluntary industry-led body established to minimise the risks of jet fuel supply disruption at eight major Australian airports and three regional international airports. The Committee comprises of all the major Australian jet fuel suppliers, and its key purposes is to communicate the status of current and future jet fuel supply availability at Australian and New Zealand major airports. It communicates through regular reporting to nominated members of government and industry.
	Intergovernmental Agreement	States and territories must be consulted prior to the declaration of a national emergency. An **Intergovernmental Agreement** (2006) sets out parameters for a co-operative response, and the National Liquid Fuel Emergency Response Plan and supporting documents describe these arrangements.
	National Liquid Fuel Emergency Response Plan	The **National Liquid Fuel Emergency Response Plan** reinforces established industry and market strategies for returning Australia to normal fuel supply levels. It includes a communication plan designed to support the successful management of any liquid fuel emergency. It can be used to bring powers under the *Liquid Fuel Emergency Act 1984* into operation if a national supply disruption is severe or may last for a long time. The Plan aims to ensure fuel is allocated in the most efficient and fair way, and that the impact of any shortage on fuel users is minimised.
	Liquid Fuel Emergency Guidelines 2019	The **Liquid Fuel Emergency Guidelines** are subordinate instrument under the *Liquid Fuel Emergency Act* 1984, which aims to ensure that both essential and non-essential users in the Australian market receive fuel in an equitable and transparent manner during an emergency, based on available supplies.
	Liquid Fuel Emergency (Activities - Essential Users) Determination 2019	The **Liquid Fuel Emergency Determination** is a subordinate instrument under the *Liquid Fuel Emergency Act 1984* which provides a predetermined list of essential user activities. This allows for greater planning and preparation, so that if a liquid fuel emergency does occur, there will be an existing strategy to prioritise the needs of these Essential Users.
	Na**tional** Oil Supply Emergency Committee Guidance Note on Essential Users	The **Guidance Note on Essential Users** sets out the current policy with respect to Essential Users in a national liquid fuel emergency, and the rationale that informs that policy. The Guidance Note also sets out some actions that fuel users can take to manage the impacts of a reduction of fuel supply. The prevailing policy is that Essential Users have been identified and the Federal Minister will be advised not to extend Essential User Status.
	Na**tional** Oil Supply Emergency Committee Guidance Note on Retail Rationing	The **Retail Rationing Framework** is part of the NLFERP and governs the way liquid fuel is rationed in Australia. The current rationing approach would involve limiting the volume of fuel a motorist can purchase up to a pre-defined daily dollar-value. A N**a**tional O**il** S**upply** E**mergency** C**ommittee Guidance Note on Retail Rationing** was agreed in 2016 to help retailers and members of

Liquid Fuels	Arrangement	Description
		the public plan for the unlikely event that retail rationing is required. All essential users are prioritised under a retail rationing framework.
	Na**tional** Oil Supply Emergency Committee Guidance Note on Jet Fuel Supply Emergencies	The **Guidance Note on Jet Fuel Supply Emergencies** was established in 2017 and describes how information on significant jet fuel supply disruptions (BLACK Traffic Light events) at National Oil Committee airports will be provided to relevant N**a**tional O**il** S**upply** E**mergency** C**ommittee** representatives and the National Oil Committee Chair. The Guidance Note supports the National Oil Committee Stakeholder Communication Protocol (2008) maintained by the National Oil Committee Chair.

TABLE 17: Liquid Fuels multi-jurisdictional and state-level joint government-industry arrangements

Appendix B: Related plans and arrangements

Document	Link
Assistance for power dependent people during power outage emergencies JSOP	Department of Health / Department of Families, Fairness and Housing
Critical Infrastructure All Sectors Resilience Annual Report (2020)	Critical Infrastructure Resilience | Emergency Management Victoria (emv.vic.gov.au)
Electricity Distribution Code 2021	Electricity Distribution Code | Essential Services Commission
Electricity Safety (Bushfire Mitigation) Regulations 2013	Electricity Safety (Bushfire Mitigation) Regulations 2013 (legislation.vic.gov.au)
Electricity Safety (Electric Line Clearance) Regulations 2020	Electricity Safety (Electric Line Clearance) Regulations 2020 (legislation.vic.gov.au)
Electricity Safety Act 1998	Electricity Safety Act 1998 (legislation.vic.gov.au)
Electricity Statement of Opportunities	AEMO | NEM Electricity Statement of Opportunities (ESOO)
Emergency Management Act 2013	Emergency Management Act 2013
Environment Protection Act 2017	Environment Protection Act 2017 (legislation.vic.gov.au)
Fuel Emergency Act 1977	Fuel Emergency Act 1977 (legislation.vic.gov.au)
Gas Industry Act 2001	Gas Industry Act 2001 (legislation.vic.gov.au)
Gas Safety Act 1997	Gas Safety Act 1997 (legislation.vic.gov.au)
Liquid Fuel Emergency (Activities—Essential Users) Determination 2019	Liquid Fuel Emergency (Activities—Essential Users) Determination 2019
Liquid Fuel Emergency Act 1984 (Cth)	Liquid Fuel Emergency Act 1984 (legislation.gov.au)
National Electricity Law	South Australian Legislation
National Electricity Rules	National Electricity Rules | AEMC
National Liquid Fuel Emergency Response Protocol	National Oil Supplies Emergency Committee (NOSEC) | Australian Institute of Petroleum (aip.com.au)
NOSEC Guidance Note Essential Users under the Liquid Fuel Emergency Act 1984	Guide Note on Essential Users (environment.gov.au)
Occupational Health and Safety Regulations 2017 - Part 5.2 Major Hazard Facilities	Occupational Health and Safety Regulations 2017 (legislation.vic.gov.au)
Pipelines Act 2005	Pipelines Act 2005 (legislation.vic.gov.au)
Power System Emergency Management Plan	AEMO | National role
Powerline Replacement Fund	Powerline replacement (energy.vic.gov.au)
Single Industry Spokesperson Protocol for Electricity in Victoria (2019)	AEMO-Single-Industry-Spokesperson-Protocol-VIC-Electricity.pdf

OFFICIAL

Single Industry Spokesperson Protocol for Gas in Victoria (2019)	2D58AADB8B04406CA2E3A10D2C89C744.ashx (aemo.com.au)
State Emergency Management Plan	State Emergency Management Plan (SEMP) | Emergency Management Victoria (emv.vic.gov.au)
Victorian Critical Infrastructure Resilience Strategy	Critical Infrastructure Resilience | Emergency Management Victoria (emv.vic.gov.au)

Appendix C: Acronyms

Acronym	Definition
AEMO	Australian Energy Market Operator
CAOiC	Control Agency Officer in Charge *(Secretary-DEECA)*
CFA	Country Fire Authority
DEECA	Department of Energy, Environment and Climate Action
ESV	Energy Safe Victoria
FRV	Fire Rescue Victoria
LOR	Lack of Reserve
SCC	State Control Centre
SCC-Energy Unit	State Control Centre – Energy Intelligence Unit
SC-E	Class 2 State Controller – Energy

OFFICIAL

deeca.vic.gov.au

OFFICIAL